



MODULE 1: PRICING AND VALUATION OF FORWARD COMMITMENTS

LEARNING OUTCOMES

- 1.a.** Describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value
- 1.b.** Describe the carry arbitrage model without underlying cashflows and with underlying cashflows
- 1.c.** Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value
- 1.d.** Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value
- 1.e.** Describe how interest rate swaps are priced, and calculate and interpret their no-arbitrage value
- 1.f.** Describe how currency swaps are priced, and calculate and interpret their no-arbitrage value
- 1.g.** Describe how equity swaps are priced, and calculate and interpret their no-arbitrage value

[LOS 1.a] Describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Recall: Forward commitments

Forward commitments

Forwards

A promise to buy or sell an asset at a future date at a price agreed to at the contract's initiation.

Futures

Similar to forwards but there is a daily settling of gains and losses by a clearinghouse.

Swaps

A promise to undertake a transaction at a set price or rate at several dates in the future.

[LOS 1.a] Describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Principles of Arbitrage-Free Pricing and Valuation

2.1. Pricing versus Valuation

Pricing

Determine the price or rate when initiating the forward commitment contract.

Valuation

Determine the value of the forward commitment, typically after it has been initiated.

2.2. No-arbitrage principles in pricing and valuation

Law of one price

Two investments with equivalent future cash flows should have the same current price

Value of a portfolio = Sum of the values of each instrument held in the portfolio **(Value additivity)**

If violated, there is arbitrage opportunity

Pricing and valuation forward based on the no-arbitrage approach must follow the law of one price

[LOS 1.a] *Describe* how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.**Principles of Arbitrage-Free Pricing and Valuation*****2.3. Key assumptions*****1**

Replicating instruments are identifiable and investable

2

Market frictions are nil

3

Short selling is allowed with full use of proceeds

4

Borrowing and lending are available at a known risk-free rate

[LOS 1.a] Describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value

3.

Pricing and Valuing Generic Forward and Futures Contracts

Carry arbitrage model

Carry arbitrage model

No-arbitrage approach in which the underlying instrument is either bought or sold along with establishing a forward position.

Account for

Carry costs

Carry benefits

Financing, storage,
insurance

Value of forward

Pricing and valuation: Details

Notation

S

Spot price of the
underlying instrument

V

Value of forward

F

Forward price

v

Value of futures

f

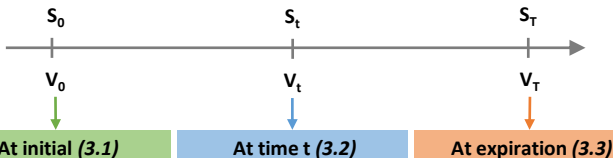
Future price

[LOS 1.a] Describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value

3.

Pricing and Valuing Generic Forward and Futures Contracts

Pricing and valuation: Details



3.1. At initial

- F_0 and f_0 are established to make the contract value zero for both the long and short parties.

$$V_0 = v_0 = 0$$

- The forward (delivery) price does not change during the life of the contract.

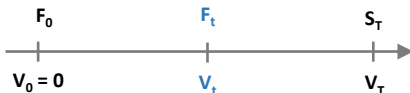
[LOS 1.a] Describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value

3.

Pricing and Valuing Generic Forward and Futures Contracts

Pricing and valuation: Details

2.2. At time t



Initial contract:
 $F_0 = \$100$

Pay \$100

Newly created
contract: $F_t = \$98$

Negative V_t

Pay \$98 instead of \$100

Newly created
contract: $F_t = \$102$

Positive V_t

Pay \$102 instead of \$100

[LOS 1.a] Describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value

3.**Pricing and Valuing Generic Forward and Futures Contracts*****Pricing and valuation: Details*****3.3. At expiration**

As we approach expiration, the price of a newly created forward or futures contract will approach the price of a spot transaction.

Convergence property: $F_T = f_T = S_T$

If the convergence property does not hold, arbitrage will force the prices to be equal at contract expiration.

[LOS 1.a] Describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value

3.

Pricing and Valuing Generic Forward and Futures Contracts

Pricing and valuation: Details

3.3. At expiration

$$F_0 = \$100$$

$$S_T = \$103$$

$$V_0 = 0$$

$$V_T = S_T - F_0$$



Investor A

Long a 6-year
forward contract

At expiration, he buys the underlying
with at \$100 instead of \$103

$$\rightarrow V_T = S_T - F_0 = \$3 \text{ (Gain)}$$

Similarly, the value of the short forward position is $V_T = F_0 - S_T$

With futures contracts

Futures contract values
are settled by margining
at the end of each trading
day when the contract is
marked-to-market

Before MTM: $v_t = f_t - f_{t-1}$

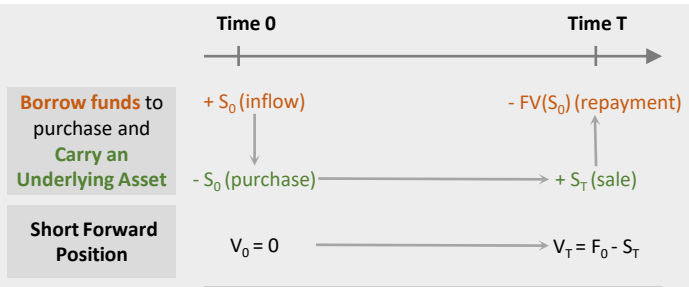
After MTM: $v_t = 0$

Gains and losses
accumulate in
the futures
traders' margin
accounts

[LOS 1.b] Describe the carry arbitrage model without underlying cashflows and with underlying cashflows

1.

Carry arbitrage model



To prevent arbitrage, the overall portfolio should have zero value at time T

Long Asset + Borrowed Funds + Short Forward

Overall Position

$$S_0 - S_0 + V_0 = 0$$

$$+S_T - FV(S_0) + V_T = 0$$

$$+S_T - FV(S_0) + (F_0 - S_T) = 0$$

$$+F_0 - FV(S_0) = 0$$

Net

0

$$F_0 = FV(S_0)$$

[LOS 1.b] *Describe* the carry arbitrage model without underlying cashflows and with underlying cashflows

1.

Carry arbitrage model

The quoted forward price does not directly reflect expectations of future underlying prices.

**FV(S_0) is
affected by**

Current price (S_0)

Interest rate

Time to expiration

Absence of arbitrage

Here we assume that forward contracts and futures contracts are priced in the same way under carry arbitrage model.

Continuous compounding (r_c)

$$F_0 = S_0 \times e^{rT}$$

Periodic compounding (r)

$$F_0 = S_0 \times (1+r)^T$$

[LOS 1.b] *Describe* the carry arbitrage model without underlying cashflows and with underlying cashflows

1.

Carry arbitrage model



Knowledge check

***Example:* Equity Futures Contract Price with Periodic Compounded Interest Rates**

Consider a 3-month forward contract on a stock currently priced at \$500, without paying any dividend during the contract life-time and suppose that the annual risk-free rate is 6%. Determine the price of the forward contract under the no-arbitrage principle.

Answer:

$$T = \frac{3}{12} = 0.25$$

$$F_0 = S_0 \times (1 + R_f)^T = \$500 \times 1.06^{0.25} = \$507.34$$

[LOS 1.b] *Describe* the carry arbitrage model without underlying cashflows and with underlying cashflows

2.

Pricing and valuing equity forward and futures

2.1. Without underlying cash flows

Unless $F_0 = FV(S_0)$, there is an arbitrage opportunity

$$F_0 > FV(S_0)$$

$$F_0 < FV(S_0)$$

Forward contract is overvalued

Forward contract is undervalued

Sell forward contract

Buy forward contract

[LOS 1.a,b] Describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value; **Describe** the carry arbitrage model without underlying cashflows and with underlying cashflows

2. Pricing and valuing equity forward and futures

2.1. Without underlying cash flows

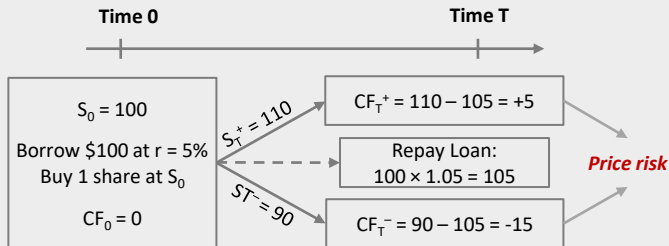


Illustration: Pricing equity forward

Given cash flows from carrying the underlying, a non-dividend-paying stock, assuming $S_0 = 100$, $r = 5\%$, and $T = 1$. Stock price at expiration might go down to $S_T^- = 90$ or up to $S_T^+ = 110$. At time 0, we have:

Case 1: Cash Flows for Long Financial Position

- Purchase one unit of the underlying at time 0 (an outflow).
- Borrow the purchase price at the risk-free rate (an inflow).



**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.1. Without underlying cash flows



Illustration: Pricing equity forward

**Case 2: Cash Flow for Long Financial Position with Short Forward contract
(to eliminate price risk)**

Strategy:

- Borrow the purchase price at the risk-free rate (an inflow).
- Purchase one unit of the underlying at time 0 (an outflow).

Since this transaction has no risk, we need the forward price (F_0) computed following the no-arbitrage approach:

$$F_0 = S_0(1 + r)^T = 100(1 + 0.05)^1 = 105.$$

- Sell a forward at $F_0 = 105$. For a short forward contract, F_0 is the price agreed to at time 0 to sell the asset at Time T.

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

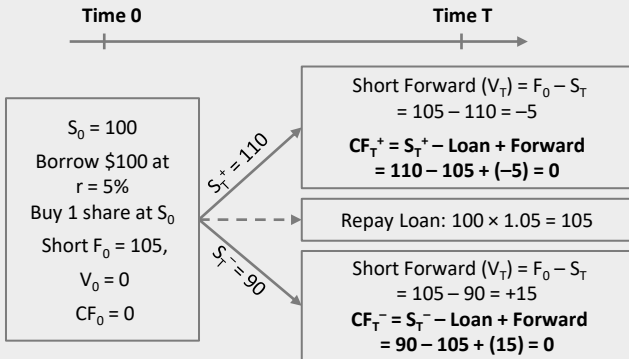
Pricing and valuing equity forward and futures

2.1. Without underlying cash flows



Illustration: Pricing equity forward

**Case 2: Cash Flow for Long Financial Position with Short Forward contract
(to eliminate price risk)**



→ There is zero net cash flow at expiration **under all circumstances.**

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.1. Without underlying cash flows

Without market frictions, arbitrage may be possible when mispricing occurs, or when $F_0 \neq FV(S_0)$.



Illustration: Arbitrage mechanics

General rule: Buy low sell high

**Scenario 1: Forward Price Greater Than Carry Arbitrage Model Price,
 $F_0 > FV(S_0)$ ($106 > 105$)**

Strategy:

- Sell the forward contract on the underlying.
- Borrow the funds to purchase the underlying.
- Purchase the underlying.

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

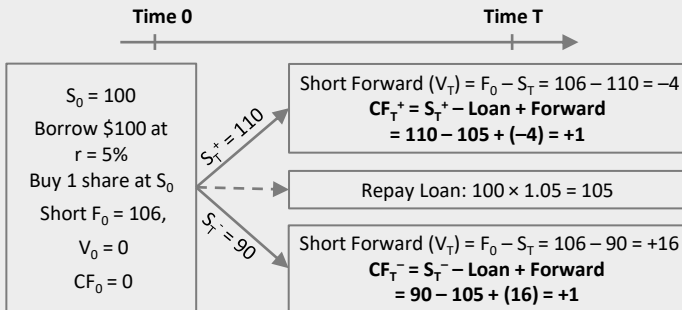
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Pricing and valuing equity forward and futures

2.1. Without underlying cash flows



Illustration: Arbitrage mechanics



At time T, we earn an arbitrage profit of +1 (under every circumstance)
→ considered risk-free

Will be pursued until forward prices fall and eliminate the arbitrage opportunity

Carry arbitrage (Carrying the underlying)

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.1. Without underlying cash flows



Illustration: Arbitrage mechanics

General rule: Buy low sell high

**Scenario 2: Cash Flow with Forward Price Lower Than Carry Arbitrage
Model Price $F_0 < FV(S_0)$ ($104 < 105$)**

Strategy:

- Buy the forward contract on the underlying.
- Sell the underlying short.
- Lend the short sale proceeds.

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

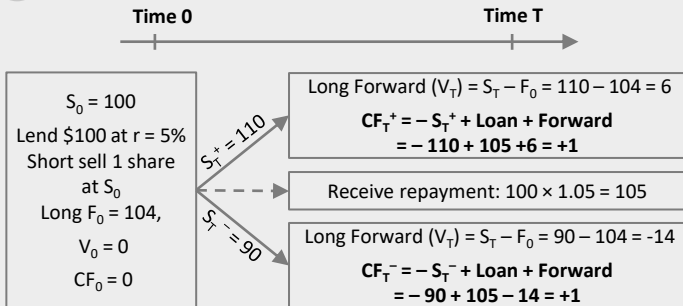
2.

Pricing and valuing equity forward and futures

2.1. Without underlying cash flows



Illustration: Arbitrage mechanics



At time T, we earn an arbitrage profit of +1 (under every circumstance)

→ Considered risk-free

Will be pursued until forward prices fall and eliminate the arbitrage opportunity

Reverse carry arbitrage (Selling short the underlying)

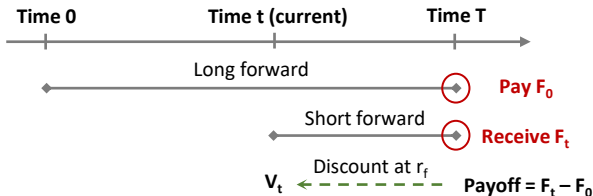
**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.1. Without underlying cash flows

Valuing equity forward and futures



V_t (long) = Present value of the payoff at time T

$$V_t = PV [F_t - F_0] = \frac{[F_t - F_0]}{(1 + r)^{T-t}}$$

$$V_t = S_t - PV [F_0] = S_t - \frac{F_0}{(1 + r)^{T-t}}$$

**[LOS 1.a,b] *Describe* how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.1. Without underlying cash flows

Valuing equity forward and futures



Knowledge check

***Example:* Forward Contract Value**

Assume that at Time 0 we entered into a one-year long forward contract with price $F_0 = 105$. Nine months later, at Time $t = 0.75$, the observed price of the underlying stock is $S_{0.75} = 110$ and the interest rate is 5%. Calculate the value of the existing forward contract expiring in three months.

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.1. Without underlying cash flows

Valuing equity forward and futures



Knowledge check

Example: Forward Contract Value

Answer:

Note that, $S_{0.75} = 110$, $r = 5\%$, and $T - t = 0.25$.

Step 1: Calculate forward price based on no-arbitrage approach at time t

Therefore, the three-month forward price at Time t is equal to

$$F_t = FV(S_t) = 110(1 + 0.05)^{0.25} = 111.3499.$$

Step 2: Calculate the difference in forward price (F_0 and F_t)

$$\text{Payoff} = F_t - F_0 = 111.35 - 105 = 6.35$$

Step 3: Calculate the present value of the different in forward price or using different in spot price at time t and PV of F_0

$$V_t = PV[F_t - F_0] = 6.35 / (1 + 0.05)^{0.25} = 6.2729.$$

Alternatively,

$$V_t = S_t - PV[F_0] = 110 - [105 / (1 + 0.05)^{0.25}] = 6.2729.$$

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.2. With underlying cash flows

Carry arbitrage model is a no-arbitrage approach in which the underlying instrument is either bought or sold along with establishing a forward position (refer to slide 5).

Carry arbitrage model accounts for:

Carry benefits (CB)

Dividends, foreign currency interest, coupon payments

Carry costs (CC)

Waste, storage, and insurance

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

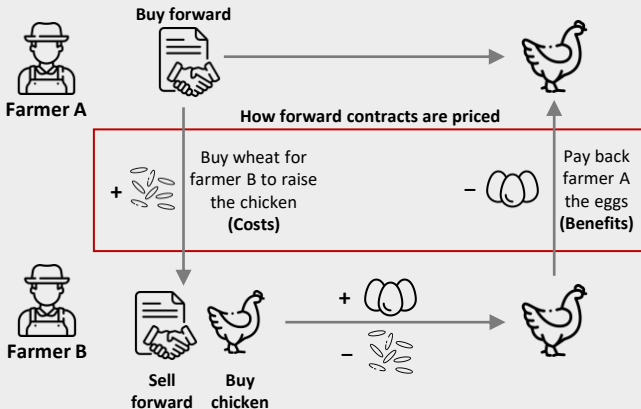
Pricing and valuing equity forward and futures

2.2. With underlying cash flows



Illustration: Carry costs and benefits

Imagine farmer A was buying a forward contract on a chicken, farmer B was selling a forward contract on a chicken. For 3 months forward contract, the chicken laid 3 eggs and ate 1kg wheat.



**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.2. With underlying cash flows

Pricing equity forward and futures with underlying cash flows

With annual compounding

Cost of carry model

$$F_0 = FV[S_0 + CC_0 - CB_0]$$

For a stock paying a dividend (D), a benefit, prior to maturity of the forward contract, forward contract price (F_0):

$$F_0 = FV[S_0 - PV(D)] = FV(S_0) - FV(D)$$



Knowledge check

Example: Spot and Forward Prices and Rates

A US stock paying a \$10 dividend in two months is trading at \$1,000. Assume the US interest rate is 5% with annual compounding.

Based on the current stock price and the no-arbitrage approach, calculate the equilibrium three-month forward price?

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

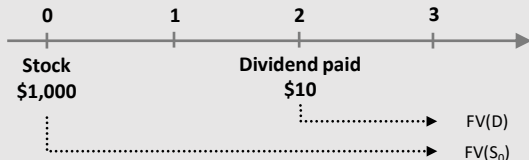
2.

Pricing and valuing equity forward and futures



Knowledge check

Example: Spot and Forward Prices and Rates



Based on the information given, we know $S_0 = \$1,000$, $r = 5\%$ (annual compounding), and $T = 0.25$. After 2 months, we will receive the benefit of a \$10 dividend, which earns interest for 1 month.

$$F_0 = FV(S_0) - FV(D) = 1,000(1 + 0.05)^{3/12} - 10(1 + 0.05)^{1/12}$$

$$= \$1,012.2722 - \$10.0407 = \$1,002.2315.$$

Or $CC_0 = 0$, and CB_0 is the PV of the dividend at time 0

$$= 10/(1 + 0.05)^{2/12} = \$9.919.$$

$$F_0 = FV(S_0 + CC_0 - CB_0) = FV(1,000 + 0 - 9.919)$$

$$= (990.081) \times (1 + 0.05)^{3/12} = \$1,002.23$$

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.2. With underlying cash flows

Pricing equity forward and futures with underlying cash flows

With continuous compounding

With stock indexes, it is difficult to account for the numerous dividend payments paid by underlying stocks that vary in timing and amount.

→ Future value of the underlying adjusted for carry costs and benefits:

$$F_0 = S_0 e^{(r_c + CC - CB)T}$$

**Dividend
index point**

A measure of the quantity of dividends attributable to a particular index.

Useful measure of the
amount of dividends paid

Useful number for
arbitrage trading

**[LOS 1.a,b] *Describe* how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.2. With underlying cash flows

Pricing equity forward and futures with underlying cash flows

With continuous compounding



Knowledge check

***Example:* Equity Futures Contract Price with Continuously Compounded Interest Rates**

The continuously compounded dividend yield on the EURO STOXX 50 is 3%, and the current stock index level is 3,500. The continuously compounded annual interest rate is 0.15%. Based on the carry arbitrage model, calculate the three-month futures price.

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.2. With underlying cash flows

Pricing equity forward and futures with underlying cash flows

With continuous compounding



Knowledge check

Example: Equity Futures Contract Price with Continuously Compounded Interest Rates

Answer:

Based on the carry arbitrage model, the futures price is

$$f_0 = S_0 e^{(r_c + CC - CB) T}$$

We assume the carry costs (CC) are 0 for a financial asset, such as a stock index. The carry benefit (CB), in this case a 3% continuous dividend yield, is greater than the financing cost r_c (0.15%), so the futures price will be below the spot price. The futures price, the future value of the underlying adjusted for carry (i.e., the dividend payments, over the next 3-months) is:

$$f_0 = 3,500 e^{(0.0015 + 0 - 0.03) \left(\frac{3}{12}\right)} = 3,475.15$$

**[LOS 1.a,b] Describe how equity forwards and futures are priced...;
Describe the carry arbitrage model...**

2.

Pricing and valuing equity forward and futures

2.2. With underlying cash flows

Valuing equity forward and futures with underlying cash flows

When the underlying has carry
benefits or carry costs

New forward price (F_t), and
initial price (F_0), are adjusted
for these benefits and costs

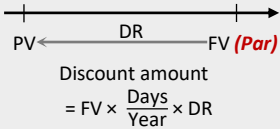
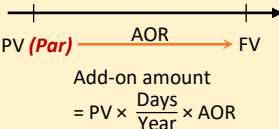
V_t = Present value of the difference in forward prices adjusted for carry
benefits and costs
= $PV[F_t - F_0]$

[LOS 1.c] Describe how interest rate forwards and futures are priced and calculate and interpret their no-arbitrage value

2.

Interest Rate Forward and Futures Contracts


 Recall: *Discount basis vs. Add-on basis*

	Discount rate (DR) basis	Add-on rate (AOR) basis
Principle	 Discount amount $= FV \times \frac{\text{Days}}{\text{Year}} \times DR$	 Add-on amount $= PV \times \frac{\text{Days}}{\text{Year}} \times AOR$
Pricing model	$PV = FV \times (1 - \frac{\text{Days}}{\text{Year}} \times DR)$	$FV = PV \times (1 + \frac{\text{Days}}{\text{Year}} \times AOR)$
Yield measure	$DR = \frac{\text{Year}}{\text{Days}} \times \frac{FV - PV}{FV}$	$AOR = \frac{\text{Year}}{\text{Days}} \times \frac{FV - PV}{PV}$

MRR market is quoted at 360-day add-on basis.

[LOS 1.c] *Describe* how interest rate forwards and futures are priced and calculate and interpret their no-arbitrage value

2.

Interest Rate Forward and Futures Contracts

Notation

L_m MRR spot rate (set at time $t = 0$) for an m -day deposit

NA Notional amount, quantity of funds initially deposited

NTD Number of total days in a year, used for interest calculations

t_m Accrual period, fraction of year for an m -day deposit— $t_m = m/NTD$

TA Terminal amount, quantity of funds repaid when the MRR deposit is withdrawn

[LOS 1.c] *Describe* how interest rate forwards and futures are priced and calculate and interpret their no-arbitrage value

2.

Interest Rate Forward and Futures Contracts



Illustration

Example: Terminal amount and Interest payment

Suppose we are considering a 90-day Eurodollar deposit ($m = 90$). Dollar MRR is quoted at 2%; thus, $L_{90} = 0.02$. If \$50,000 is initially deposited, then $NA = \$50,000$. MRR is stated on an actual over 360-day count basis (often denoted ACT/360) with interest paid on an add-on basis.

Hence, $t_m = 90/360 = 0.25$. Accordingly, the terminal amount can be expressed as:

$$TA = NA \times [1 + L_m t_m]$$

$$\text{Interest paid} = TA - NA = NA \times [L_m t_m]$$

$$TA = \$50,000 \times [1 + 0.02(90/360)] = \$50,250$$

$$\text{Interest} = \$50,250 - \$50,000 = \$250$$

[LOS 1.c] Describe how interest rate forwards and futures are priced and calculate and interpret their no-arbitrage value

2.

Interest Rate Forward and Futures Contracts

Forward rate agreement (FRA)

Forward rate agreement (FRA)

FRA is an over-the-counter (OTC) forward contract in which the underlying is an interest rate on a deposit.

Long FRA



Fixed-rate payer
Floating-rate receiver

Suited for a firm
planning to borrow
at fixed rate in the
future

**Hedge against
rising rates**

Pay fixed rate

Pay floating rate

Short FRA



Fixed-rate receiver
Floating-rate payer

Bank or financial
institution hoping to
lock in a fixed lending
rate in the future

**Hedge against
falling rates**

[LOS 1.c] Describe how interest rate forwards and futures are priced and calculate and interpret their no-arbitrage value

2.

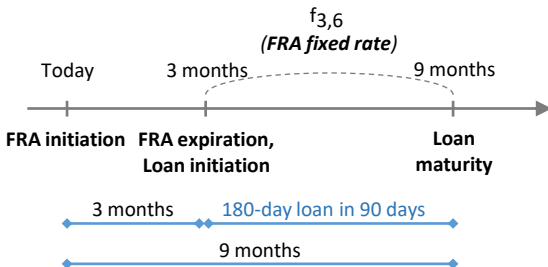
Interest Rate Forward and Futures Contracts

Forward rate agreement (FRA)

FRA price

Fixed interest rate so that the FRA value is zero on the initiation date to eliminate arbitrage opportunities (no initial exchange of cash flows).

How do you interpret a 3x9 FRA? (on a 30/360 basis)



FRA fixed rate: The rate on a deposit begins in 3 months and ends in 9 months (last for 6 months).

Long 3x9 FRA = Long a nine-month MRR deposit and short a three-month MRR deposit.

[LOS 1.c] Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

FRA settlement

There are two ways to settle at expiration:

FRA expiration

Loan maturity

Interest rate is set

Interest payment is made

**Advanced set,
settled in arrears**

Interest payment is made

FRA payment is discounted back

**Used in swaps and
interest rate options**

**Advanced set,
advanced settle**

Used in FRAs

Settlement amount (long FRA)

$$NA \times \{[L_m - FRA_0]t_m\} / [1 + D_m t_m]$$

Settlement amount (short FRA)

$$NA \times \{[FRA_0 - L_m]t_m\} / [1 + D_m t_m]$$

- It is often assumed at FRA expiration that $D_m = L_m$*
- D_m is the discount rate for the FRA settlement*

[LOS 1.c] *Describe* how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

FRA settlement



Knowledge check

Example: Calculating Interest on MRR Spot and FRA Payment

In 30 days, a UK company expects to make a bank deposit of £10,000,000 for a period of 90 days at 90-day MRR set 30 days from today. The company is concerned about a possible decrease in interest rates. Its financial adviser suggests that it negotiate today a 1×4 FRA, an instrument that expires in 30 days and is based on 90-day MRR. The company enters a £10,000,000 notional amount 1×4 receive-fixed FRA that is advanced set, advanced settled (note the company is the short-side of this FRA contract). The appropriate discount rate for the FRA settlement cash flows is 2.40%. After 30 days, 90-day MRR in British pounds is 2.55%.

1. Calculate the interest actually paid at maturity on the UK company's bank deposit.
2. If the FRA was initially priced so that $FRA_0 = 2.60\%$, calculate the payment received to settle it.
3. If the FRA was initially priced so that $FRA_0 = 2.50\%$, calculate the payment received to settle it.

[LOS 1.c] Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

FRA settlement



Knowledge check

Example: Calculating Interest on MRR Spot and FRA Payment

Answer:

1. The interest actually paid at maturity on the UK company's bank deposit:

Since $m = 90$, we use $L90 = 2.55\%$.

Therefore,

$$TA = 10,000,000 \times [1 + 0.0255(0.25)] = £10,063,750.$$

So, the interest paid at maturity is £63,750.

[LOS 1.c] Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

FRA settlement



Knowledge check

Example: Calculating Interest on MRR Spot and FRA Payment

Answer:

2. If the FRA was initially priced so that $FRA_0 = 2.60\%$, the payment received to settle it will be:

The settlement amount of the 1×4 FRA at expiration for receive-fixed (the short) is:

$$\begin{aligned} & NA \times \{[FRA_0 - L_m] t_m\} / [1 + D_m t_m] \\ &= 10,000,000 \times \{[0.0260 - 0.0255](0.25)\} / [1 + 0.0240(0.25)] \\ &= \text{£}1,242.54. \end{aligned}$$

Since the short FRA involves paying floating, the short benefited from a decline in rates. Note D_m does not equal L_m in this example.

[LOS 1.c] *Describe* how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

FRA settlement



Knowledge check

Example: Calculating Interest on MRR Spot and FRA Payment

Answer:

3. If the FRA was initially priced so that $FRA_0 = 2.50\%$, the payment received to settle it will be:

The data are similar to those in the previous question, but the initial FRA rate is now 2.50% and not 2.60%. Thus, the settlement amount of the 1×4 FRA at expiration for receive-fixed (the short) is:

$$\begin{aligned} & NA \times \{[FRA_0 - L_m]t_m\} / [1 + D_m t_m] \\ &= 10,000,000 \times \{[0.0250 - 0.0255](0.25)\} / [1 + 0.0240(0.25)] \\ &= -£1,242.54. \end{aligned}$$

The short-side in the FRA suffered from a rise in rates because it is paying floating.

[LOS 1.c] Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

Pricing FRAs

The FRA price is the **implied forward rate** for the period beginning when the FRA expires to the underlying loan maturity.

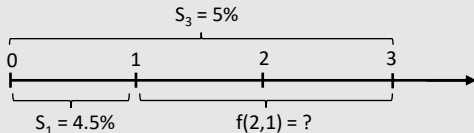
→ Pricing an FRA is like pricing a forward contract.



Recall: Calculate forward rate from spot rate

Calculate the 2-year forward rate 1 year from today, given 1-year spot rate is 4.5% and 3-year spot rate is 5%?

Answer:



$$(1 + S_1)^1 [1 + f(2,1)]^2 = (1 + S_3)^3$$

$$(1 + 4.5\%)^1 [1 + f(2,1)]^2 = (1 + 5\%)^3$$

$$f(2,1) = \sqrt{(1 + 5\%)^3 / (1 + 4.5\%)^1} - 1 = 5.25\%$$

[LOS 1.c] Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

Pricing FRAs

The FRA price is the **implied forward rate** for the period beginning when the FRA expires to the underlying loan maturity.



Knowledge check

Example: Calculating the price of an FRA

Calculate the price of a 1×4 FRA (i.e., a 90-day loan, 30 days from now). The current 30-day MRR is 4% and 120-day MRR is 5%.

Answer:

The actual (unannualized) rate on the 30-day loan is:

$$L_{30} = 4\% \times \frac{30}{360} = 0.333\%$$

The actual (unannualized) rate on the 120-day loan is:

$$L_{120} = 5\% \times \frac{120}{360} = 1.667\%$$

[LOS 1.c] Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

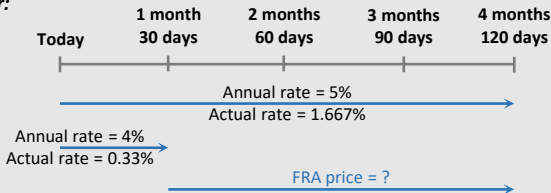
Forward rate agreement (FRA)

Pricing FRAs



Knowledge check

Answer:



We wish to calculate the actual rate on a 90-day loan from day 30 to day 120:

$$\text{Price of a 1 x 4 FRA} = \frac{1 + L_{120}}{1 + L_{30}} - 1 = \frac{1 + 1.01667}{1 + 1.0333} - 1 = 1.33\%$$

We can annualize this rate as: $1.33\% \times \frac{360}{90} = 5.32\%$

This is the no-arbitrage forward rate — the forward rate that will make the values of the long and the short positions in the FRA both zero at the initiation of the contract.

[LOS 1.c] Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

Valuing an FRA at Maturity

- In the previous example the long in the FRA has the “right” to borrow money 30 days from inception for a period of 90 days at the forward rate.
 - If interest rates increase, the long will profit as the contract has fixed a borrowing rate below the now-current market rate.
- To value the FRA we need to take ***the present value of the interest savings at the end of the loan term.***



Knowledge check

Example: Calculating the value of an FRA at maturity (i.e., cash payment at settlement)

Continuing the prior example for a 1×4 FRA, assume a notional principal of \$1 million. Suppose that, at contract expiration, the 90-day rate has increased to 6%, which is above the contract rate of 5.32%. Calculate the value of the FRA at maturity, which is equal to the cash payment at settlement.

[LOS 1.c] Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

Valuing an FRA at Maturity



Knowledge check

Example: Calculating the value of an FRA at maturity (i.e., cash payment at settlement)

Answer:

Step 1: Calculate the interest savings at the end of the loan term

The interest savings at the end of the loan term (compared to the market rate of 6%) will be:

$$\left[\left(0.06 \times \frac{90}{360} \right) - \left(0.0532 \times \frac{90}{360} \right) \right] \times \$1,000,000 = \$1,700$$

Step 2: Calculate the present value of the savings at FRA settlement date

The present value of this amount at the FRA settlement date (90 days prior to the end of the loan term) discounted at the current rate of 6% is:

$$\frac{\$1,700}{1 + \left(0.06 \times \frac{90}{360} \right)} = \$1,674.88$$

[LOS 1.c] Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

Valuing an FRA at Maturity

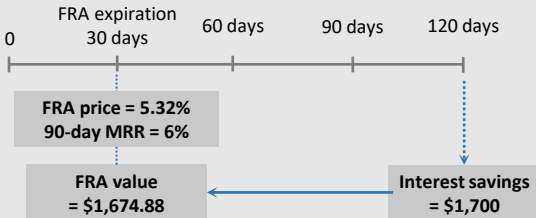


Knowledge check

Example: Calculating the value of an FRA at maturity (i.e., cash payment at settlement)

Answer:

This will be the cash settlement payment from the short to the long at the expiration of the contract. Note that we have discounted the savings in interest at the end of the loan term by the market rate of 6% that prevails at the contract settlement date for a 90-day term, as shown in the following figure.



[LOS 1.c] *Describe* how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

Valuing an FRA before Maturity

The calculation of the “savings” on the loan will be the same as in our previous example, except that we need to use the “new” FRA price that would be quoted on a contract covering the same period as the original “loan”.



Knowledge check

Example: Calculating value of an FRA before settlement

Value a 5.32% 1×4 FRA with a principal amount of \$1 million 10 days after initiation if 110-day MRR is 5.9% and 20-day MRR is 5.7%.

Answer:

Step 1: Find the “new” FRA price on a 90-day loan 20 days from today. This is the current 90-day forward rate at the settlement date, 20 days from now.

$$\left[\frac{1 + \left(0.059 \times \frac{110}{360} \right)}{1 + \left(0.057 \times \frac{20}{360} \right)} - 1 \right] \times \frac{360}{90} = 0.0592568$$

[LOS 1.c] Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Forward rate agreement (FRA)

Valuing an FRA before Maturity



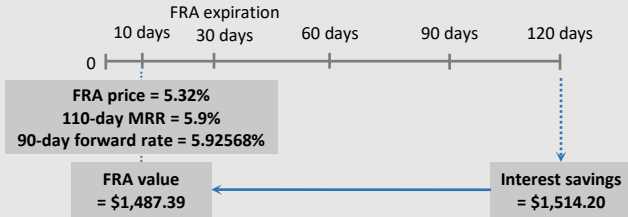
Knowledge check

Step 2: Calculate the interest difference on a \$1 million, 90-day loan made 20 days from now at the forward rate calculated previously compared to the FRA rate of 5.32%.

$$\left[\left(0.0592568 \times \frac{90}{360} \right) - \left(0.0532 \times \frac{90}{360} \right) \right] \times \$1,000,000 = \$1,514.20$$

Step 3: Discount this amount at the current 110-day rate.

$$\frac{\$1,514.20}{1 + \left(0.059 \times \frac{110}{360} \right)} = \$1,487.39$$



[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Fixed-income forward and futures contracts characteristics

Fixed-income forward and futures contracts have several unique issues that influence the specifics of the carry arbitrage model.

Unique characteristics of Fixed-income forward and futures contracts

Fixed-income securities prices are quoted without accrued interest **(1.1)**

Fixed-income futures contracts often have more than one bond that can be delivered by the seller **(1.2)**

When multiple bonds can be delivered for a particular futures contract, a cheapest-to-deliver (CTD) bond typically emerges after adjusting for the conversion factor **(1.3)**

[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Fixed-income forward and futures contracts characteristics

1.1. Quoted bond price and accrued interest

**Quoted price /
Clean price**



Accrued interest



**Full price /
Dirty price**

$$\begin{aligned} &\text{Accrued interest} \\ &= \text{Accrual period} \times \text{Periodic coupon amount} \end{aligned}$$

$$\begin{aligned} &\textbf{Accrued interest} \\ &AI = (NAD/NTD) \times (C/n) \end{aligned}$$

Where:

- **NAD:** number of accrued days since the last coupon payment
- **NTD:** number of total days during the coupon payment period
- **N:** number of coupon payments per year
- **C:** stated annual coupon amount

For example, after two months (60 days), a 3% semi-annual coupon bond with par of 1,000 would have accrued interest of $AI = (60/180) \times (30/2) = 5$

[LOS 1.d] *Describe* how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Fixed-income forward and futures contracts characteristics

1.2. Conversion factor (CF)

Bonds trade at different prices based on maturity and stated coupon

Conversion factor (CF) is used to make all deliverable bonds approximately **equal in price** when settling a futures contract

1.3. Cheapest-to-deliver bond (CTD bond)

If there are several candidates for delivery

The **least expensive** for the seller to purchase in the open market to settle the obligation will be delivered

[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Pricing fixed-income forward and futures

Bonds quoted with accrued interest

The futures or forward price is simply the future value of the underlying in which finance costs, carry costs, and carry benefits are all incorporated.

$$F_0 = \text{Future value of underlying adjusted for carry cash flows} \\ = FV(S_0 + CC_0 - CB_0)$$

Bonds quoted without accrued interest

Let AI_0 denote the accrued interest at Time 0, using the same equation, we have:

$$F_0 = \text{Future value of underlying adjusted for carry cash flows} \\ = FV(S_0 + CC_0 - CB_0)$$

In which:

$$S_0 = \text{Quoted bond price} + \text{Accrued interest} = B_0 + AI_0$$

**The quoted bond price plus the accrued interest is the spot price: It is in fact the price you would have to pay to buy the bond.*

[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Pricing fixed-income forward and futures

Bonds quoted without accrued interest

- The carry benefits are the bond's fixed coupon payments: $CB_0 = PVCI$ (present value of all coupon interest paid over the forward contract horizon from Time 0 to Time T)
- There are no carry costs: $CC = 0$

$$\begin{aligned} F_0 &= FV(S_0 + CC_0 - CB_0) \\ &= FV[S_0 + 0 - PVCI] \\ &= FV[B_0 + AI_0 - PVCI] \end{aligned}$$

*The fixed-income forward or futures price **including the conversion factor**, termed the “adjusted price,” can be expressed as:*

$$\begin{aligned} F_0 &= Q_0 \times CF \\ \Leftrightarrow Q_0 &= F_0 / CF \end{aligned}$$

- F_0 : future price
- Q_0 : the quoted futures price
- CF: conversion factor

[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Pricing fixed-income forward and futures

Total profit or loss on a long position in fixed-income futures at expiration (Time T) is the quoted bond price minus the initial futures price.

$$v_T = B_T - F_0$$

$$v_T = (S_T - AI_T) - F_0$$

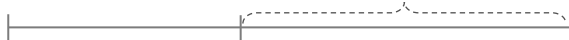
Timeline for Bond Futures and Forwards

Time 0, Forward or
Futures Contract
Initiation

Time T, Forward or
Futures Contract
Expiration

Time Y, Time
to Maturity of
Bond at Time T

Time T + Y
Underlying
Bond Matures



Quoted bond price	B_0
Accrued Interest	AI_0
Spot Bond price	$S_0 = B_0 + AI_0$
Quoted forward or futures price	Q_0
Forward of Futures Adjusted price	$F_0 = Q_0 \times CF$

Quoted bond price	B_T
Accrued Interest	AI_T
Spot Bond price	$S_T = B_T + AI_T$
Profit on Long Forward or Futures	$V_T = B_T - F_0$ $= (S_T - AI_T) - F_0$
Profit on Short Forward or Futures	$V_T = F_0 - B_T$ $= F_0 - (S_T - AI_T)$

[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

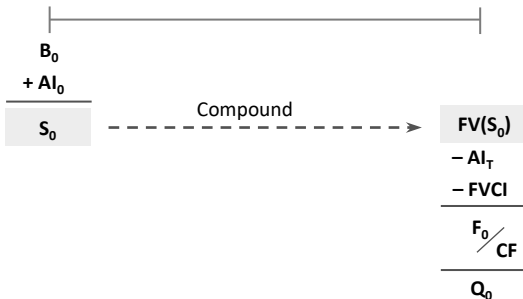
Pricing fixed-income forward and futures

To eliminate arbitrage profit, or $v_T = 0$

$$\begin{aligned}
 v_T &= (S_T - AI_T) - F_0 = 0 \\
 &\Leftrightarrow F_0 = S_T - AI_T \\
 &\Leftrightarrow F_0 + AI_T = S_T = F_T = FV[B_0 + AI_0 - PVCI] \\
 &\Leftrightarrow F_0 = FV(S_0) - AI_T - FVCI
 \end{aligned}$$

Time 0, Forward or
Futures Contract
Initiation

Time T, Forward or
Futures Contract
Expiration



[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Pricing fixed-income forward and futures

Illustration: *Disequilibrium and Arbitrage Opportunity*

Assume we observe a 3-month forward contract, so $T = 0.25$, on a bond that expires at some time in the future, $T + Y$, and this bond is currently quoted (B_0) at 107% of par. There are no coupon payments for this bond over the life of the forward contract, so $PVCI = 0.0$.

Bond

Quoted Bond Price	B_0	107.00
PV of Coupon Interest	$PVCI$	0
Accrued Interest at Time 0	AI_0	0.07
Accrued Interest at Time T	AI_T	0.20

Futures

Quoted Futures Price	Q_0	135.00
Conversion Factor	CF	0.80
Adjusted Futures Price	$F_0 (= Q_0 \times CF)$	108.00

Interest Rate

For Discounting/Compounding	r	0.20%
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[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Pricing fixed-income forward and futures

 Illustration: Disequilibrium and Arbitrage Opportunity

We have:

Full spot price of the bond is: $S_0 = B_0 + AI_0 = 107 + 0.07 = 107.07$

Futures price (F_0), which is the future value adjusted for carry cash flows (using Equation $F_0 = FV[B_0 + AI_0 - PVCIF]$), is:

$$F_0 = (107 + 0.07 - 0)(1.002)^{0.25} = 107.12$$

Adjusted futures price using the quoted futures price ($Q_0 = 135$) and the conversion factor ($CF = 0.80$) including accrued interest is:

$$F_0 + AI_T = Q_0 \times CF + AI_T = 135 \times 0.8 + 0.2 = 108.20$$

Here, the adjusted futures price (including accrued interest) is 108.20, while the cost to buy and carry the bonds is 107.12 $\rightarrow$ futures contract is overpriced by $(108.2 - 107.12) = 1.08 \rightarrow$ **there is an arbitrage opportunity.**

We would simultaneously:

1. Sell the overpriced futures contract
2. Borrow funds to purchase the bonds
3. Buy the underpriced deliverable bonds

$\rightarrow$ Arbitrage profit at time 0 = $1.08(1.002)^{-0.25} = 1.0795$, or = 1.08 at time T

[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Pricing fixed-income forward and futures

Illustration: *Disequilibrium and Arbitrage Opportunity*

Or using Equation $F_0 = FV(S_0) - AI_T - FVCI$, we have:

Adjusted futures price, $F_0 = 108$

$$FV(S_0) - AI_T - FVCI = 107.12 - 0.2 - 0 = 106.92$$

→ Arbitrage profit at time T = $108 - 106.92 = 1.08$

Quoted futures price (Q_0):

$$\begin{aligned} Q_0 &= [1/CF] \{FV[B_0 + AI_0] - AI_T - FVCI\} \\ &= (1/0.8) \{(1 + 0.002)0.25(107 + 0.07) - 0.20 - 0.0\} = 133.65. \end{aligned}$$

Recall, a futures price of 135 was used as the quoted price, Q_0 . Any quoted futures price higher than the equilibrium futures price of 133.65 will present arbitrage opportunities.

[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Pricing fixed-income forward and futures



Knowledge check

Example: Estimating the Euro-Bund Futures Price

Euro-bund futures have a contract value of €100,000, and the underlying consists of long-term German debt instruments with 8.5 to 10.5 years to maturity. They are traded on the Eurex. Suppose the underlying 2% coupon (semi-annual payment) German bund is quoted at €108 and has accrued interest of €0.083 (15 days since last coupon paid). The euro-bund futures contract matures in one month (30 days). At contract expiration, the underlying bund will have accrued interest of €0.25; there are no coupon payments due until after the futures contract expires; and the current one-month risk-free rate is 0.1%. The conversion factor is 0.729535.

In this case, we have the following:

$$T = 1/12, CF = 0.729535, B_0 = 108, FVCI = 0,$$

$$AI_0 = (15/180 \times 2\%/2) = €0.083, AI_T = (45/180 \times 2\%/2) = €0.25,$$

and $r = 0.1\%$.

Calculate the equilibrium euro-bund quoted futures price (Q_0) based on the carry arbitrage model.

[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

1.

Pricing fixed-income forward and futures



Knowledge check

Example: Estimating the Euro-Bund Futures Price

Answer:

The equilibrium euro-bund futures price can be found using the carry arbitrage model:

$$Q_0 = [1/CF]\{FV[B_0 + AI_0] - AI_T - FVCI\}$$

$$= [1/0.729535] [(1 + 0.001)^{1/12}(108 + 0.083) - 0.25 - 0] = 147.82.$$

In equilibrium, the quoted euro-bund futures price should be approximately €147.82 based on the carry arbitrage model.

[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Valuing fixed-income forward

Without daily settlement, the value of a forward is not formally realized until expiration.

The net cash flow at maturity is the difference in the price at which we sold, F_t , and the price we agreed to pay, F_0 , or $(F_t - F_0)$.

Since the position is riskless, the value to the long at time t should be

$V_t = \text{Present value of difference in forward prices at time } t = \text{PV} [F_t - F_0]$



Knowledge check

Example: Estimating the Value of a Euro-Bund Forward Position

Suppose that one month ago, we purchased five euro-bund forward contracts with two months to expiration and a contract notional value of €100,000 each at a price of 145 (quoted as a percentage of par). The euro-bund forward contract now has one month to expiration. The current annualized one-month risk-free rate is 0.1%. Based on the current forward price of 148, calculate the value of the euro-bund forward position.

[LOS 1.d] Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

2.

Valuing fixed-income forward

The net cash flow at maturity is the difference in the price at which we sold, F_t , and the price we agreed to pay, F_0 , or $(F_t - F_0)$.

Since the position is riskless, the value to the long at time t should be

$V_t = \text{Present value of difference in forward prices at time } t = \text{PV} [F_t - F_0]$



Knowledge check

Example: Estimating the Value of a Euro-Bund Forward Position

Answer:

Because we are given both forward prices, the solution is simply the present value of the difference in forward prices at expiration.

$$V_t = \text{PV}[F_t - F_0] = (148 - 145)/(1 + 0.001)^{1/12} = 2.99975.$$

This is 2.9997 per €100 par value because this forward price was quoted as a percentage of par. Because five contracts each with €100,000 par were entered, we have $0.029997(\text{€}100,000)5 = \text{€}14,998.75$.

[LOS 1.e] Describe how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

Overview: Swap contracts

Definition

An agreement to exchange (or swap) a series of cash flows at certain periodic dates.

Market participants often use swaps to transform one series of cash flows into another.

**Types of
swaps**

Interest rate swaps (LOS 1.e)

Currency swaps (LOS 1.f)

Equity swaps (LOS 1.g)

We will be pricing and valuing each type of swap contract in the next few Learning outcomes.

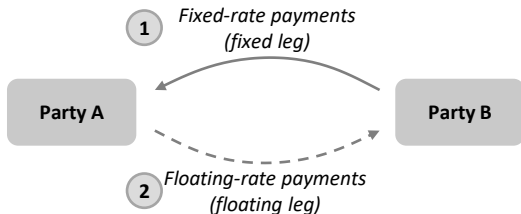
[LOS 1.e] Describe how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

1.

Recall: Interest rate swap

Definition

An interest rate swap is an agreement between two parties to exchange one stream of **interest payments** for another, over a set period of time.



1 Fixed leg cash flow (FS) with the fixed rate (r_{FIX})

$$FS = AP \times r_{FIX}$$

2 Floating leg cash flow (S_i) with floating rate during period i ($r_{FLT,i}$)

$$S_i = AP \times r_{FLT,i}$$

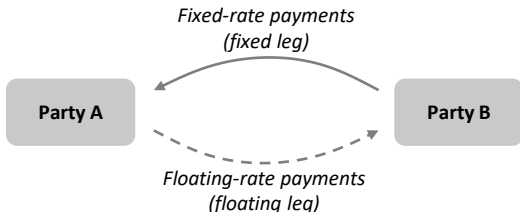
$AP = \text{days}/n$ is the accrual period.

For simplicity, we assume that the notional amounts are equal to one ($NA = 1$).

[LOS 1.e] Describe how interest rate swaps are priced, and calculate and interpret their no-arbitrage value

1.

Recall: Interest rate swaps



Net cash flow for party A

Receive fixed: + FS

Pay floating: - S_i

$$FS - S_i = AP \times (r_{FIX} - r_{FLT,i})$$

Net cash flow for party B

Receive floating: + S_i

Pay fixed: - FS

$$S_i - FS = AP \times (r_{FLT,i} - r_{FIX})$$

[LOS 1.e] *Describe* how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing an Interest rate swap

Pricing follows no-arbitrage principle: If we want to price a swap, we have to see if the counterparty can engage in arbitrage to mitigate it.

Establish a portfolio with similar cash flow to the initial swap

Counterparty may enter an opposite portfolio

If the portfolio values are unequal → Arbitrage opportunity

Equate the swaps to prevent arbitrage

[LOS 1.e] Describe how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing an Interest rate swap



Illustration: Pricing a swap

1. Establish a swap as a replicating portfolio

Party A

Receives fixed-rate payments

Pays floating-rate payments

Long fixed rate bond

Short floating rate bond

2. Counterparty may establish an opposite portfolio

Party B

Receives floating-rate payments

Pays fixed-rate payments

Long floating-rate bond

Short sell fixed-rate bond

[LOS 1.e] Describe how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing an Interest rate swap



Illustration: Pricing a swap

3. Identify the portfolio cash flows and values

Position	Step	Time 0	Time 1	...	Time n
Swap	Receive-fixed, pay-floating swap	V_{swap}	$FS - S_1$	...	$FS - S_n$
Offsetting portfolio	Buy floating-rate bond	V_{FLT}	$+ S_1$	...	$+ S_n$
	Short-sell fixed-rate bond	V_{FIX}	$- FS$	...	$- FS$
Net cash flows		$V_{\text{swap}} = -V_{\text{FLT}} + V_{\text{FIX}}$	0	0	0

[LOS 1.e] Describe how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing an Interest rate swap



Illustration: Pricing a swap

4. To prevent arbitrage, equate the portfolio values

$$V_{\text{swap}} = -V_{\text{FLT}} + V_{\text{FIX}} = 0$$

Interpretation:

- The value of a receive-fixed, pay-floating interest rate swap is simply the value of *buying a fixed-rate bond* and *issuing a floating-rate bond*.
- The bond values are the PVs of all the expected interest and par payments, thus we have the signs preceding the values.

Pricing the swap: Determine the fixed rate r_{FIX} so that the value of the swap at initiation is zero.

[LOS 1.e] Describe how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing an Interest rate swap

Pricing the swap: Determine the fixed rate so that the value of the swap at initiation is zero.

Value of fixed-rate bond

$$V_{\text{FIX}} = r_{\text{FIX}} \times \sum_{i=1}^n DF_i + DF_n$$

- r_{FIX} is the coupon amount of the fixed bond.
- The bond has par value of 1.

Value of floating-rate bond

$$V_{\text{FLT}} = 1$$

At reset date, Coupon rate is set to match the Discount rate
→ Bond sells at par

$$V_{\text{swap}} = -V_{\text{FLT}} + V_{\text{FIX}} = 0$$

$$\rightarrow r_{\text{FIX}} \times \sum_{i=1}^n DF_i + DF_n = 1$$

The value of the fixed bond must equal the value of the floating bond

Pricing an interest rate swap

$$r_{\text{FIX}} = \frac{1 - DF_n}{\sum_{i=1}^n DF_i}$$

[LOS 1.e] Describe how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing an Interest rate swap



Knowledge check

Example: Fixed Swap Rate Based on Present Value Factors

Suppose we are pricing a five-year MRR-based interest rate swap with annual resets (30/360 day count). The estimated present value factors are given in the following table.

Maturity	Discount factors
1	0.990099
2	0.977876
3	0.965136
4	0.951529
5	0.937467

Calculate the fixed rate of the swap.

[LOS 1.e] Describe how interest rate swaps are priced, and **calculate** and **interpret** their no-arbitrage value

2.

Pricing an Interest rate swap



Knowledge check

Example: Fixed Swap Rate Based on Present Value Factors

Recall:

$$r_{\text{FIX}} = \frac{1 - DF_n}{\sum_{i=1}^n DF_i}$$

$$\sum_{i=1}^n DF_i = 0.990099 + 0.977876 + 0.965136 + 0.951529 + 0.937467 = 4.822107$$

$$r_{\text{FIX}} = \frac{1 - 0.937467}{4.822107} = 0.012968$$

[LOS 1.e] Describe how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

3.

Valuation of an Interest rate swap

After initiation, the term structure of interest rates will likely imply a different swap rate, $r_{FIX,t}$. On the other hand, the price of entering a new swap has changed.

General idea: How much are we better off by entering an interest rate swap in the past compared to now?

Identify the cash flows of initial swap

Establish a new offsetting swap

Identify the net cash flows of the swaps in subsequent periods

Value is the PVs of the difference between the cash flows

[LOS 1.e] Describe how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

3.

Valuation of an Interest rate swap



Illustration: *Valuing a swap*

1. Establishing the value ($V_{\text{swap},0}$) and cash flows of the initial swap

Position	Step	Time 0	...	Time t+1	Time n
Swap	Receive-fixed, pay-floating swap	$V_{\text{swap},0}$	...	$FS_0 - S_{t+1}$	$FS_0 - S_n$

FS_0 is the fixed payments based on initial swap rate.

2. Establish the cash flows of a new offsetting swap

Position	Step	Time 0	Time t	Time t+1	Time n
Offsetting swap	Receive-floating, pay-fixed swap	(N/A)	(N/A)	$-FS_t + S_{t+1}$	$-FS_t + S_n$

FS_t is the fixed payments based on swap rate at the time of valuation.

[LOS 1.e] Describe how interest rate swaps are priced, and calculate and interpret their no-arbitrage value

3.

Valuation of an Interest rate swap



Illustration: Valuing a swap

3. Identify the net cash flows

Position	Step	Time 0	Time t	Time t+1	...	Time n
Swap	Receive-fixed, pay-floating swap	$V_{\text{swap},0}$	...	$FS_0 - S_{t+1}$	...	$FS_0 - S_n$
Offsetting swap	Receive-floating, pay-fixed swap	(N/A)	(N/A)	$-FS_t + S_{t+1}$	...	$-FS_t + S_n$
Net cash flows			$V_{\text{swap},t}$	$FS_0 - FS_t$	...	$FS_0 - FS_t$

[LOS 1.e] Describe how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

3.

Valuation of an Interest rate swap



Illustration: *Valuing a swap*

4. Determine the value of the swap at time t

The value of a receive-fixed swap at time t is the sum of the present values of the difference in fixed payments.

Value of a receive-fixed swap

$$\begin{aligned} V_{\text{swap},t} &= NA \times (FS_0 - FS_t) \\ &= NA \times AP(r_{\text{FIX},0} - r_{\text{FIX},t}) \times \sum_{i=1}^n DF_i \end{aligned}$$

n is the number of remaining cash flows from time t

→ The negative of this amount is the value to the fixed-rate payer.

Value of a pay-fixed swap

$$\begin{aligned} -V_{\text{swap},t} &= NA \times (-FS_0 + FS_t) \\ &= NA \times AP(-r_{\text{FIX},0} + r_{\text{FIX},t}) \times \sum_{i=1}^n DF_i \end{aligned}$$

[LOS 1.e] Describe how interest rate swaps are priced, and calculate and interpret their no-arbitrage value

3.

Valuation of an Interest rate swap

Receive-fixed, pay-floating				Pay-fixed, receive-floating			
Swap		Portfolio position		Rates	Portfolio position		Rates
Replicating portfolio	Time 0	Long fixed bond	Short floating bond	$r_{\text{FIX},0} - r_{\text{FLT},0}$	Long floating bond	Short fixed bond	$r_{\text{FLT},0} - r_{\text{FIX},0}$
Offsetting portfolio	Time t	Short fixed bond	Long floating bond	$r_{\text{FLT},t} - r_{\text{FIX},t}$	Short floating bond	Long fixed bond	$r_{\text{FIX},t} - r_{\text{FLT},t}$
Rates for Swap valuation	Time t			$r_{\text{FIX},0} - r_{\text{FIX},t}$			$r_{\text{FIX},t} - r_{\text{FIX},0}$

[LOS 1.e] *Describe* how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

3.

Valuation of an Interest rate swap



Knowledge check

Example: Receive-Fixed Swap Value Based on Present Value Factors

Suppose two years ago we entered a €100,000,000 seven-year receive-fixed MRR-based interest rate swap with annual resets. The fixed rate in the swap contract entered two years ago was 2.0%. The estimated present value factors are given in the following table.

Maturity	Discount factors
1	0.990099
2	0.977876
3	0.965136
4	0.951529
5	0.937467
Sum	4.822107

Calculate the value for the swap party receiving the fixed rate.

[LOS 1.e] *Describe* how interest rate swaps are priced, and *calculate* and *interpret* their no-arbitrage value

3.
Valuation of an Interest rate swap


Knowledge check

***Example:* Receive-Fixed Swap Value Based on Present Value Factors**

Answer:

$$r_{\text{FIX},0} = 2\%$$

$$r_{\text{FIX},t} = 1.3\%$$

$$\begin{aligned} V_{\text{swap},t} &= NA \times AP (r_{\text{FIX},0} - r_{\text{FIX},t}) \times \sum_{i=1}^n DF_i \\ &= €100,000,000 \times (0.02 - 0.013) \times 4.822107 \\ &= €3,375,000. \end{aligned}$$

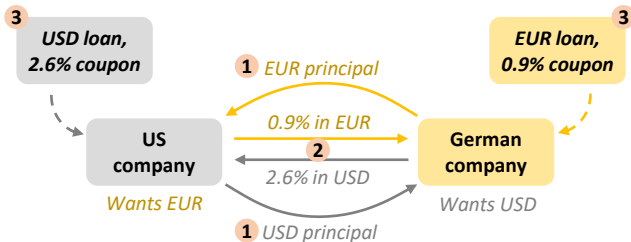
[LOS 1.f] Describe how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value

1.

Currency swaps

Definition

A contract in which two counterparties agree to exchange future interest payments in different currencies.



Key features of currency swap

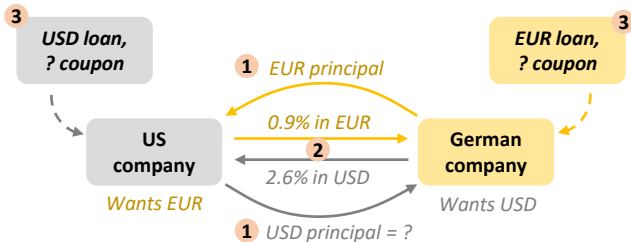
- 1 Exchange of notional amounts at both the *initiation* of the swap and at the *expiration* of the swap.
- 2 Payments on each leg is in a different currency, thus they are *not netted*.
- 3 Each leg of the swap can be *either fixed or floating*.

We will be focusing on fixed-for-fixed currency swap.

[LOS 1.f] Describe how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing a currency swap



Pricing a currency swap: Determine the following variables such that the currency swap value is zero at initiation:

- Appropriate notional amount in the other currency given one currency.
- Two fixed interest rates

We can view the swap as a pair of fixed rate bonds

The swap value is zero when both bond values are equal (valued at par)

[LOS 1.f] Describe how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing a currency swap



Illustration: Pricing a currency swap

1. Determine the fixed rate for USD bond and EUR bond

In order to prevent arbitrage, both bonds must be valued at par (1):

For USD bond:

$$V_{USD} = r_{FIX,USD} \times \sum_{i=1}^n DF_{i,USD} + DF_{n,USD} = 1$$

Fixed rate of USD bond

$$r_{FIX,USD} = \frac{1 - DF_{n,USD}}{\sum_{i=1}^n DF_{i,USD}}$$

Where:

- V_{USD} = Value of USD bond
- $r_{FIX,USD}$ = Coupon rate of USD bond

For EUR bond:

$$V_{EUR} = r_{FIX,EUR} \times \sum_{i=1}^n DF_{i,EUR} + DF_{n,EUR} = 1$$

Value of EUR bond

$$r_{FIX,EUR} = \frac{1 - DF_{n,EUR}}{\sum_{i=1}^n DF_{i,EUR}}$$

Where:

- V_{USD} = Value of EUR bond
- $r_{FIX,EUR}$ = Coupon rate of EUR bond

[LOS 1.f] Describe how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing a currency swap



Illustration: Pricing a currency swap

2. Determine the equivalent notional amount in the other currency

To prevent arbitrage, the par or principal notional amounts are set to reflect the current spot exchange rate.

$$V_{\text{USD}} = V_{\text{EUR}} \times S_0$$

Where: S_0 = Spot exchange rate (USD/EUR)

Generalization

**Fixed rate of
currency a bond**

$$r_{\text{FIX},a} = \frac{1 - DF_{n,a}}{\sum_{i=1}^n DF_{i,a}}$$

Notional amount

$$V_a = V_b \times S_0$$

**Fixed rate of
currency b bond**

$$r_{\text{FIX},b} = \frac{1 - DF_{n,b}}{\sum_{i=1}^n DF_{i,b}}$$

Note: There are two yield curves for two swap fixed rates.

[LOS 1.f] Describe how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value



Knowledge check

Example: Currency Swap Pricing with Spot Rates

A US company needs to borrow 100 million Australian dollars (A\$) for one year for its Australian subsidiary. The company decides to issue US\$-denominated bonds in an amount equivalent to A\$100 million. Then, the company enters into a one-year currency swap with quarterly reset (30/360 day count) and the exchange of notional amounts at initiation and at maturity. At the swap's initiation, the US company receives the notional amount in Australian dollars and pays to the counterparty, a swap dealer, the notional amount in US dollars. At the swap's expiration, the US company pays the notional amount in Australian dollars and receives from the counterparty the notional amount in US dollars.

Days to maturity	Discount factor (A\$1)	Discount factor (US\$1)
90	0.993789	0.999750
180	0.987167	0.999251
270	0.980152	0.998502
360	0.972763	0.997506

Assume that the counterparties in the currency swap agree to an A\$/US\$ spot exchange rate of 1.140 (expressed as number of Australian dollars for US\$1).

[LOS 1.f] Describe how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value



Knowledge check

Example: Currency Swap Pricing with Spot Rates

1. The annual fixed swap rates for Australian dollars and US dollars, respectively, will be closest to:

- A. 2.80% and 0.10%.
- B. 2.77% and 0.25%.
- C. 2.65% and 0.175%.

Answer: B

$$r_{\text{AUD}} = \frac{1 - DF_{n,\text{AUD}}}{\sum_{i=1}^n DF_{i,\text{AUD}}} = 0.00692381 \text{ or } 0.692381\%.$$

$$r_{\text{USD}} = \frac{1 - DF_{n,\text{USD}}}{\sum_{i=1}^n DF_{i,\text{USD}}} = 0.00062422 \text{ or } 0.062422\%.$$

[LOS 1.f] *Describe* how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value



Knowledge check

***Example:* Currency Swap Pricing with Spot Rates**

2. The notional amount (in US\$ millions) will be closest to:

- A. 88
- B. 100
- C. 114

Answer: A

The US dollar notional amount is calculated as A\$100 million divided by the current spot exchange rate, A\$1.140/US\$1.

From $NA_a = NA_b \times S_0$, we have $A\$100,000,000 = A\$1.14/US\$1 \times NA_b$. Solving for NA_b we have $US\$87,719,298 = A\$100,000,000/(A\$1.14/US\$1)$.

[LOS 1.f] Describe how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value

3.

Valuing a currency swap

Exchange rate fluctuation

Interest rate fluctuation

Currency swap value may change after initiation

We can view the swap as a pair of fixed rate bonds

Long bond in currency a

Short bond in currency b

Payments to be received

$$NA_a(r_{FIX,a} \times \sum_{i=1}^n DF_{i,a} + DF_{n,a})$$

Payments to be made

$$NA_b(r_{FIX,b} \times \sum_{i=1}^n DF_{i,b} + DF_{n,b})$$

Different between payments of bonds

Value of currency swap

$$V_{\text{swap}} = NA_a(r_{FIX,a} \times \sum_{i=1}^n DF_{i,a} + DF_{n,a}) - S_t \times NA_b(r_{FIX,b} \times \sum_{i=1}^n DF_{i,b} + DF_{n,b})$$

[LOS 1.f] Describe how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value



Knowledge check

Example: Currency Swap Valuation with Spot Rates

The fixed rates were found to be 2.7695% for Australian dollars and 0.2497% for US dollars. Initially, the notional amount in US dollars was determined to be US\$87,719,298 with a spot exchange rate of A\$1.14 for US\$1. Assume 60 days have passed since swap initiation and we now observe the following updated market information:

Days to maturity	Discount factor (A\$1)	Discount factor (US\$1)
30	0.998336	0.999584
120	0.993707	0.998668
210	0.989609	0.998253
300	0.986031	0.998336

The currency spot exchange rate (S_t) is now A\$1.13 for US\$1.

[LOS 1.f] Describe how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value



Knowledge check

Example: Currency Swap Valuation with Spot Rates

1. The current value to the swap dealer in A\$ of the currency swap entered 60 days ago will be closest to:

- A. −A\$13,557,000.
- B. A\$637,620.
- C. A\$2,145,200.

Answer: C

After 60 days, the new exchange rate is 1.13A\$/1US\$ and the term structure of interest rates has changed in both markets. V_{swap} is the value of the swap to the party receiving interest payments in Australian dollars, which is the swap dealer. Thus, the value to the swap dealer receiving A\$ is:

$$V_{\text{swap}} = NA_a(r_{\text{FIX},a} \times \sum_{i=1}^n DF_{i,a} + DF_{n,a}) - S_t \times NA_b(r_{\text{FIX},b} \times \sum_{i=1}^n DF_{i,b} + DF_{n,b})$$

$$= A\$100,000,000 \times [0.00692375 (3.967683) + 0.986031] - 1.13 (A\$/1US\$) \times (US\$87,719,298) \times [0.00062425 (3.994841) + 0.998336] = A\$2,145,167.$$

[LOS 1.g] Describe how equity swaps are priced, and *calculate* and *interpret* their no-arbitrage value

1.

Equity swaps

Definition

2 parties agree to exchange a series of cash flows

Party A

Pays a variable series that will be determined by an equity

Party B

Pays a variable series determined by either

Different equity

Rate

Types of equity swaps

- Receive equity return
- Pay fixed

- Receive equity return
- Pay floating

- Receive equity return
- Pay another equity return

Underlying for equity leg

Individual stock

Published stock index

Custom portfolio

The equity leg cash flows can be with or without dividends.

[LOS 1.g] Describe how equity swaps are priced, and *calculate* and *interpret* their no-arbitrage value

1.

Equity swaps

Types of swap

Receive-equity, pay-fixed

Receive-equity, pay-floating

Receive-equity, pay-equity

Cash flows

 $NA \times (\text{Equity return} - \text{Fixed rate})$
 $NA \times (\text{Equity return} - \text{Floating rate})$
 $NA \times (\text{Equity return}_a - \text{Equity return}_b)^*$

**a and b denotes different equities.*

Cash flows

 Equity leg (S_i)

$$S_i = NA_E \times R_E$$

Fixed interest rate leg (FS)

$$FS = NA_E \times AP_{\text{FIX}} \times r_{\text{FIX}}$$

Where:

- NA_E is the notional amount
- R_E is the return of the equity
- AP_{FIX} is the accrual period of the fixed leg
- r_{FIX} is the fixed rate

[LOS 1.g] *Describe* how equity swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing an equity swap

Pricing an equity swap: Follows the no arbitrage rule, similar to the previous types of swaps.

Identify the cash flows of the initial swap

Counterparty may enter an offsetting portfolio

If the portfolio values are unequal → Arbitrage opportunity

Equate the swaps to prevent arbitrage

[LOS 1.g] Describe how equity swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing an equity swap



Illustration: *Pricing an equity swap*

1. Identify the positions of the initial swap

Party A

Receives fixed

Pays equity

2. Establish an offsetting portfolio

Party B

Receives equity

Pays fixed

Buy NA_E of equity

Short sell fixed-rate bond

[LOS 1.g] Describe how equity swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing an equity swap


 Illustration: *Pricing an equity swap*

3. Identify cash flows and values of the swap and portfolio

Position	Step	Time 0	Time 1	...	Time n
Equity swap	Receive-fixed, pay-equity swap	$-V_{EQ}$	$FS - S_1$	...	$FS - S_n$
Offsetting portfolio	Buy NA_E of equity	$-NA_E$	$+S_1$	...	$S_n + NA_E$
	Short-sell fixed-rate bond	$+V_{FIX}$	$-FS$	...	$-FS - Par$
Net cash flows		$-V_{EQ} - NA_E + V_{FIX}$	0	0	$NA_E - Par$

[LOS 1.g] Describe how equity swaps are priced, and *calculate* and *interpret* their no-arbitrage value

2.

Pricing an equity swap



Illustration: *Pricing an equity swap*

4. To prevent arbitrage, an initial cash outflow is required to yield the cash inflow at time n

$$-V_{EQ} - NA_E + V_{FIX} = PV(\text{Par} - NA_E)$$

$$\rightarrow V_{EQ} = -NA_E + V_{FIX} - PV(\text{Par} - NA_E) = 0$$

$$\rightarrow V_{FIX} = NA_E + PV(\text{Par} - NA_E)$$

For the terminal cash flow to be zero, we must set the bond par value to equal the initial equity position $\rightarrow NA_E = \text{Par} = 1$

$$V_{FIX} = NA_E + PV(0) = 1$$

$$\rightarrow r_{FIX} \times \sum_{i=1}^n DF_i + DF_n = 1$$

Price of equity swap

$$r_{FIX} = \frac{1 - DF_n}{\sum_{i=1}^n DF_i}$$

[LOS 1.g] *Describe* how equity swaps are priced, and *calculate* and *interpret* their no-arbitrage value

3.

Valuing an equity swap

Valuing an equity swap: The process is similar to valuing an interest rate swap, except that we need to adjust the value of the notional amount of equity.

Value of a receive-fixed swap

$$V_{EQ,t} = V_{FIX}(C_0) - (S_t/S_{t-1}) NA_E - PV(\text{Par} - NA_E)$$

[LOS 1.f] *Describe* how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value



Knowledge check

***Example:* Equity swap valuation**

A \$10 million principal, one-year, quarterly settlement equity swap has a fixed rate of 6.05%. The index at inception is 985. After 30 days have passed, the index stands at 996 and the term structure of MRR is 6%, 6.5%, 7%, and 7.5% for terms of 60, 150, 240, and 330 days, respectively. The respective discount factors are 0.99010, 0.97363, 0.95541, and 0.93567.

Calculate the value of the swap to the fixed-rate payer on Day 30.

[LOS 1.f] Describe how currency swaps are priced, and *calculate* and *interpret* their no-arbitrage value



Knowledge check

Example: Currency Swap Valuation with Spot Rates

Answer:

Value of the fixed side (per \$1 notional) = present value of coupons + PV of principal

$$= (0.0605/4) (0.99010 + 0.97363 + 0.95541 + 0.93567) + (0.93567) (1) \\ = 0.9940$$

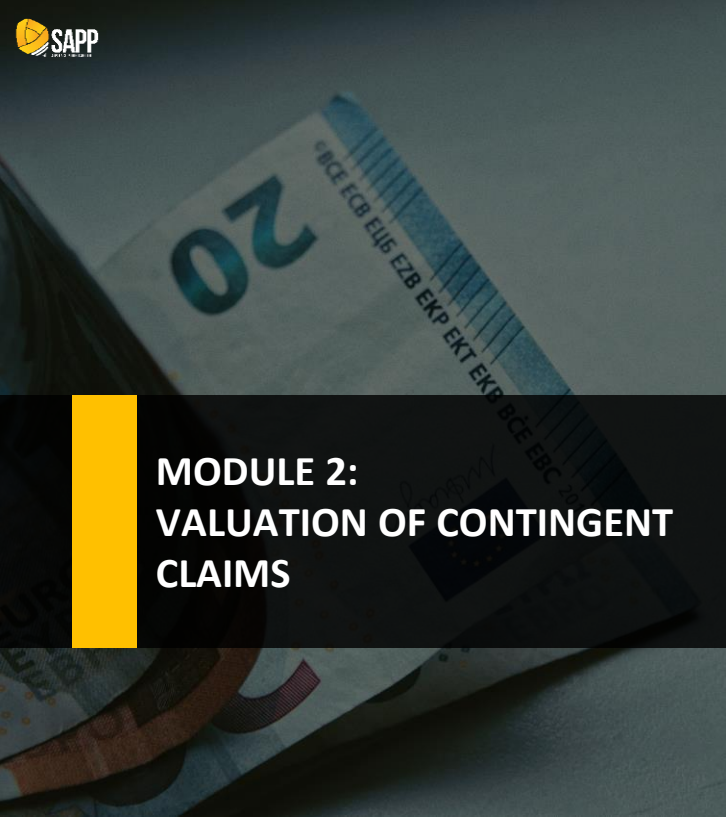
Value of the fixed-rate side of the swap (\$10 million notional) = $0.9940 \times \$10$ million = \$9,940,000

The value of \$10,000,000 invested in the index after 30 days is:

$$\$10,000,000 \times (996/985) = \$10,111,675$$

From the standpoint of the fixed-rate payer, the value of the swap after 30 days is:

$$\$10,111,675 - \$9,940,000 = \$171,675.$$



MODULE 2: VALUATION OF CONTINGENT CLAIMS

LEARNING OUTCOMES

- 2.a.** Describe and interpret the binomial option valuation model and its component terms.
- 2.b.** Describe how the value of a European option can be analyzed as the present value of the option's expected payoff at expiration.
- 2.c.** Identify an arbitrage opportunity involving options and describe the related arbitrage.
- 2.d.** Calculate the no-arbitrage values of European and American options using a two-period binomial model.
- 2.e.** Calculate and interpret the value of an interest rate option using a two-period binomial model.
- 2.f.** Identify assumptions of the Black–Scholes–Merton option valuation model
- 2.g.** Interpret the components of the Black–Scholes–Merton model as applied to call options in terms of a leveraged position in the underlying

LEARNING OUTCOMES

- 2.h.** Describe how the Black–Scholes–Merton model is used to value European options on equities and currencies
- 2.i.** Describe how the Black model is used to value European options on futures
- 2.j.** Describe how the Black model is used to value European interest rate options and European swaptions
- 2.k.** Interpret each of the option Greeks
- 2.l.** Describe how a delta hedge is executed
- 2.m.** Describe the role of gamma risk in options trading
- 2.n.** Define implied volatility and explain how it is used in options trading

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

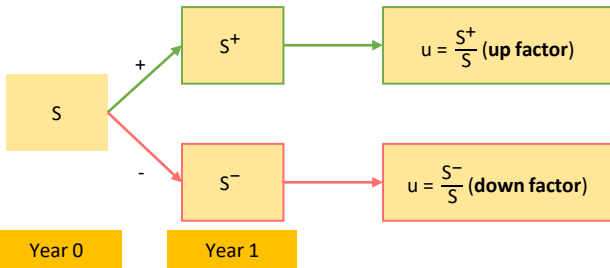
1.

One-period Binomial Model

We have been familiar with the concept of binomial tree. In this module, the two possible outcomes of the underlying is denoted S^+ and S^- for the up occurs and down occurs respectively.

One-period binomial tree

Return of movements



The value of the underlying after a movement

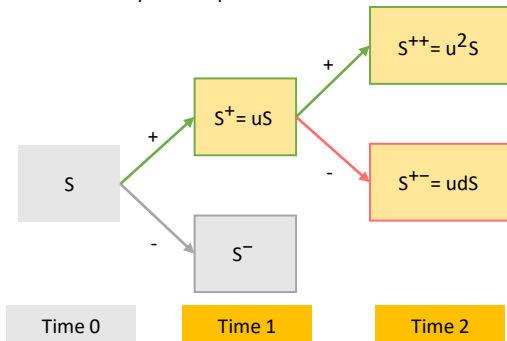
After an up-move:
 $S^+ = uS$

After a down-move:
 $S^- = dS$

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

1. One-period Binomial Model

We now look at a part of a two-period binomial model, after an up move occurred. This is basically an one-period model.



We now consider the underlying has a call option. The value measured at time 1 is denoted as c^+ . The value at expiration is either:

$$c^{++} = \text{Max}(0, S^{++} - X) = \text{Max}(0, u^2S - X), \text{ or}$$

$$c^{+-} = \text{Max}(0, S^{+-} - X) = \text{Max}(0, udS - X).$$

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

2.
How to value an option – No-arbitrage Approach

The portrait of an arbitrageur


Arbitrager

Hi, I'm an Arbitrageur. I
want to make money
**without using my money or
taking any risk**

2 Rules of being an arbitrageur

Rule # 1: Do not use your own money.

Rule # 2: Do not take any price risk.

Still make money anyways!

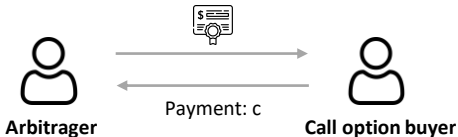
We approach our following valuation method by assuming there is **no arbitrage opportunity exist**. (Because if it exist, all the arbitrageurs will come and exploit until they no longer can, i.e. **arbitrage-free**)

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

2. How to value an option – No-arbitrage Approach

To value an option, we are taking the perspective of an arbitrageur. Note that there are no costs or benefits of owning the underlying instrument, and the valuation of call and put options follow the same framework.

Step 1: Writing a call option on the underlying stock



Arbitrager's value

After selling the call, the arbitrager received c .

Arbitrager's position

For writing the call option,

- the arbitrager will gain if the underlying goes down, and
- lose if the underlying go up

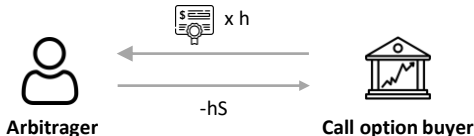
→ The arbitrager is taking a short position in the stock.

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

2. How to value an option – No-arbitrage Approach

We recall that by being an arbitrager, **we take no risk (rule #2 of being an arbitrager)**. Therefore, to offset the risk of writing a call option (equal short the stock, we need to take a long position by buying the stock.

Step 2: Buying h share of stock



Arbitrager's value

To buy h shares of stock, the arbitrager pays hS (S is the price of one share). Total received/paid by the arbitrager now is: $+c - hS$

Arbitrager's position

Arbitrager is now in a risk-free position, after writing the call and buying h shares.

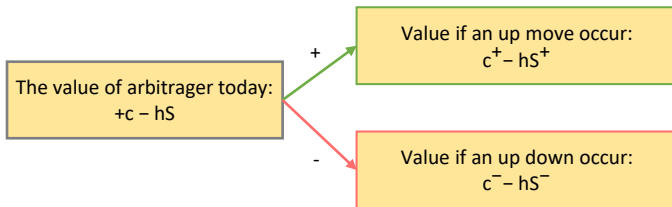
[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

2.

How to value an option – No-arbitrage Approach

Hedge ratio: How much is h ?

Obviously, we can not just purchasing any amount of shares. The right amount of share to offset the position of writing a call option is call **the hedge ratio** (therefore it is denoted h).



In time step 1, the arbitrageur exit the position. The cash flow then would be:

- $-c^+ + hS^+$ if the up move occurred, or;
- $-c^- + hS^-$ if the down move occurred.

The arbitrageur do not have to payout any money:

$$-c^- + hS^- = -c^+ + hS^+$$

We solve for h :

$$h = \frac{c^+ - c^-}{S^+ - S^-} \geq 0$$

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

2.

How to value an option – No-arbitrage Approach

Example: Calculate hedge ratio

A non-dividend-paying stock is currently trading at €100. A call option has one year to mature, the periodically compounded risk-free interest rate is 5.15%, and the exercise price is €100. Assume a single-period binomial option valuation model, where $u = 1.35$ and $d = 0.74$.

Calculate the optimal hedge ratio.

Answer

Step 1: First, we calculate the value of up and down move after one year

$$S^+ = uS = 1.35(100) = 135$$

$$S^- = dS = 0.74(100) = 74$$

Step 2: With the values after one year, we can decide the value of the call option

$$c^+ = \text{Max}(0, uS - X) = \text{Max}(0, 135 - 100) = 35$$

$$c^- = \text{Max}(0, dS - X) = \text{Max}(0, 74 - 100) = 0$$

Step 3: Use the hedge ratio formula

$$h = \frac{c^+ - c^-}{S^+ - S^-} = \frac{35 - 0}{135 - 74} = 0.573770$$

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

2. How to value an option – No-arbitrage Approach

By being an arbitrageur, **we don't spend our money (rule #1 of being an arbitrageur)**. Thus, we will fund our position by lending or borrowing.

Step 3: Funding the position by borrowing or lending

Time 0

Discount

Time 1

Funding needed:
 $PV(-c^+ + hS^+)$

Cash flow if an up move occur:
 $-c^+ + hS^+$

Funding needed:
 $PV(-c^- + hS^-)$

Cash flow if an up down occur:
 $-c^- + hS^-$

Initial cash outlay

$$+c - hS + PV(-c^- + hS^-) = 0, \text{ or}$$

$$+c - hS + PV(-c^+ + hS^+) =$$

Call option's value

$$c = hS + PV(c^- - hS^-), \text{ or}$$

$$c = hS + PV(c^+ - hS^+)$$

Put option's value

$$p = hS + PV(p^- - hS^-), \text{ or}$$

$$p = hS + PV(p^+ - hS^+)$$

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

2.

How to value an option – No-arbitrage Approach

Initial cashflow for arbitrageur

Writing the call option: $+c$

Buying h shares of stock: $-hS$

Funding (lending or borrowing):
 $PV(-c^- + hS^-) = PV(-c^+ + hS^+)$

Initial cash outlay

$$+c - hS + PV(-c^- + hS^-) = 0, \text{ or}$$

$$+c - hS + PV(-c^+ + hS^+) = 0$$

Call option's value

$$c = hS + PV(c^- - hS^-), \text{ or}$$

$$c = hS + PV(c^+ - hS^+)$$

Put option's value

$$p = hS + PV(p^- - hS^-), \text{ or}$$

$$p = hS + PV(p^+ - hS^+)$$

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

3.

How to value an option – Expectation Approach

The **expectation approach** results in an identical value as the no-arbitrage approach, but it is easier to compute. The formula are viewed as follows:

The probability of an up move: $\pi = [FV(1) - d] / (u - d)$ with $FV(1) = 1 + r$

Call option's value

$$c = PV[\pi c^+ + (1 - \pi) c^-]$$

Put option's value

$$p = PV[\pi p^+ + (1 - \pi) p^-]$$

Terminal call option payoffs

$$E(c_1) = \pi c^+ + (1 - \pi) c^-$$

Terminal put option payoffs

$$E(p_1) = \pi p^+ + (1 - \pi) p^-$$

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

3.

How to value an option – Expectation Approach

Example: Single-Period Binomial Call Value

A non-dividend-paying stock is currently trading at €100. A call option has one year to mature, the periodically compounded risk-free interest rate is 5.15%, and the exercise price is €100. Assume a single-period binomial option valuation model, where $u = 1.35$ and $d = 0.74$.

Calculate the call option value

Answer

$$\pi = [FV(1) - d] / (u - d) = (1.0515 - 0.74) / (1.35 - 0.74) = 0.510656,$$

$$\text{where } FV(1) = (1 + r) = 1.0515.$$

Thus the call value by the expectations approach is

$$c = PV[\pi c^+ + (1 - \pi) c^-] = 0.951022[(0.510656)35 + (1 - 0.510656)0] = \text{€}16.998,$$

$$\text{where } PV(1) = 1 / (1 + r) = 1 / (1.0515) = 0.951022.$$

Note that the call value by the no-arbitrage approach yields the same answer:

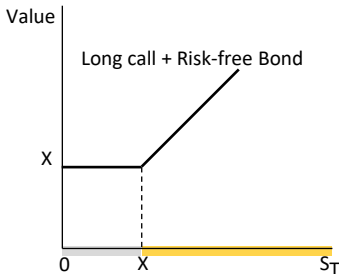
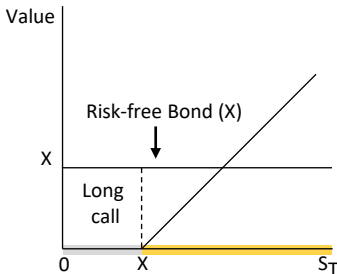
$$c = hS + PV(c^- - hS^-) = 0.573770(100) + 0.951022[-0.573770(74) + 0] = \text{€}16.998.$$

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

4.
Put-call parity
Fiduciary call

An option strategy that involve:

- long a call option with exercise price X (C_0),
- buy a zero-coupon bond pays X at maturity ($PV(X)$).



Call option not exercised, $V = 0$

Call option exercised, $V = S_T - X$

The combination value:

- $(S_T - X) + X = S_T$ if the call option is exercised;
- $0 + X = X$ if the call option is not exercised.

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

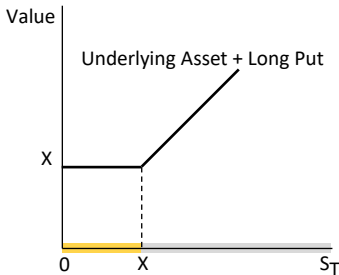
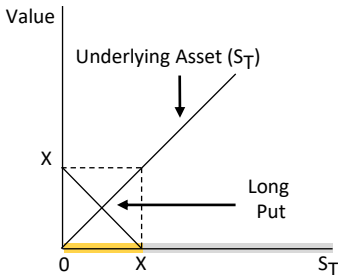
4.

Put-call parity

Protective put

An option strategy that involve:

- long a put option with exercise price X (P_0),
- buy the underlying stock (S_0).



Put option not exercised, $V = 0$

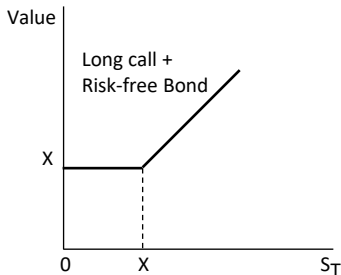
Put option exercised, $V = X - S_T$

The combination value:

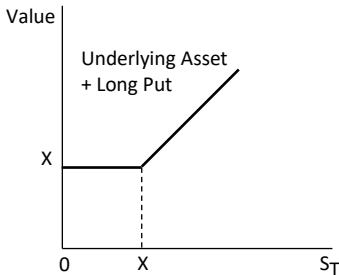
- $(X - S_T) + S_T = X$ if the put option is exercised;
- $0 + S_T = S_T$ if the put option is not exercised.

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

4. Put-call parity



Fiduciary Call



Protective Put

The value of **fiduciary call** and **protective put** is the same in all time. Therefore, we have the put-call parity relation:

Put-call parity

$$S_0 + P_0 = C_0 + PV(X)$$

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

4.

Put-call parity

Example: Using put-call parity

A 1-year call option on the common stock of Cross Reef, Inc., with an exercise price of \$60 is trading for \$8. The current stock price is \$62. The risk-free rate is 4%. Calculate the price of the put option implied by put-call parity.

Answer

According to put-call parity, to prevent arbitrage, the price of the put option must be:

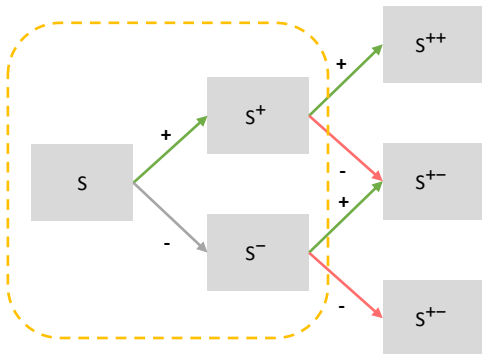
$$S_0 + P_0 = C_0 + PV(X)$$

$$\begin{aligned} P_0 &= C_0 - S_0 \frac{X}{(1 + R_f)^T} \\ &= \$8 - \$62 + \frac{\$60}{1.04} \\ &= \$3.69 \end{aligned}$$

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

5.

Valuing An Option Using Two-Period Binomial Model



Two-Period Binomial Lattice as Three One-Period Binomial Lattices

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

5.

Valuing An Option Using Two-Period Binomial Model

Step 1: Calculate the stock values at the end of two periods.

Step 2: Calculate the 3 possible option payoffs at the end of two periods.

Step 3: Calculate the expected option payoff at the end of two periods ($t = 2$) using the up- and down-move probabilities.

Step 4: Discount the expected option payoff ($t = 2$) back one period at the risk-free rate to find the option values at the end of the first period ($t = 1$).

Step 5: Calculate the expected option value at the end of one period ($t = 1$) using up- and down-move probabilities.

Step 6: Discount the expected option value at the end of one period ($t = 1$) back one period at the risk-free rate to find the option value today ($t = 0$)

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

5.
Valuing An Option Using Two-Period Binomial Model
Example: Valuing a call option on a stock with a two-period model

Suppose you own a stock currently priced at \$50 and that a two-period European call option on the stock is available with a strike price of \$45. The up-move factor is 1.25 and the down-move factor is 0.80. The risk-free rate per period is 7%. Compute the value of the call option using a two-period binomial model.

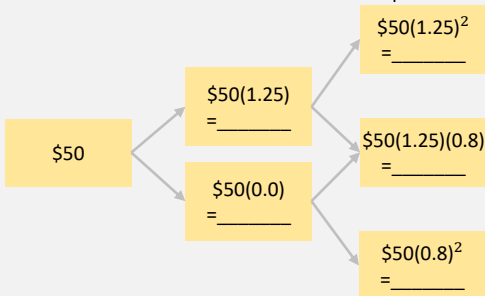
Answer

The probability of an up-move and a down-move:

$$\pi = [(1+r) - d]/(u - d) = (1 + 0.07 - 0.8)/(1.25 - 0.8) = 0.6$$

$$1 - \pi = 1 - 0.6 = 0.4$$

Step 1: Calculate the stock values at the end of two periods.



[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

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Valuing An Option Using Two-Period Binomial Model

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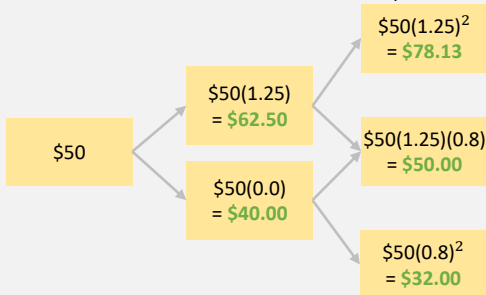
Answer

The probability of an up-move and a down-move:

$$\pi = [(1+r) - d]/(u - d) = (1 + 0.07 - 0.8)/(1.25 - 0.8) = 0.6$$

$$1 - \pi = 1 - 0.6 = 0.4$$

Step 1: Calculate the stock values at the end of two periods.



[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

5.
Valuing An Option Using Two-Period Binomial Model

Step 2: Calculate the 3 possible option payoffs at the end of two periods.

$$c^{++} = \text{Max}(0, S^{++} - X) = \text{Max}(0, u^2 S - X) = \text{max}(0, \underline{\hspace{1cm}} - \$45) = \underline{\hspace{1cm}}$$

$$c^{+-} = \text{Max}(0, S^{+-} - X) = \text{Max}(0, udS - X) = \text{max}(0, \underline{\hspace{1cm}} - \$45) = \underline{\hspace{1cm}}$$

$$c^{--} = \text{Max}(0, S^{--} - X) = \text{Max}(0, d^2 S - X) = \text{max}(0, \underline{\hspace{1cm}} - \$45) = \underline{\hspace{1cm}}$$

Step 3: Calculate the expected option payoff at the end of two periods ($t = 2$) using the up- and down-move probabilities.

$$E(c_2 | 1+) = \pi c^{++} + (1 - \pi) c^{+-} = (0.6) \underline{\hspace{1cm}} + (0.4) \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$E(c_2 | 1-) = \pi c^{+-} + (1 - \pi) c^{--} = (0.6) \underline{\hspace{1cm}} + (0.4) \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

Step 4: Discount the expected value back to time 1

$$c^+ = \frac{E(c_2 | 1+)}{1 + R_f} = \underline{\hspace{1cm}}$$

$$c^- = \frac{E(c_2 | 1-)}{1 + R_f} = \underline{\hspace{1cm}}$$

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

5.

Valuing An Option Using Two-Period Binomial Model

Step 2: Calculate the 3 possible option payoffs at the end of two periods.

$$c^{++} = \text{Max}(0, S^{++} - X) = \text{Max}(0, u^2 S - X) = \text{max}(0, \$78.13 - \$45) = \$33.13$$

$$c^{+-} = \text{Max}(0, S^{+-} - X) = \text{Max}(0, udS - X) = \text{max}(0, \$50.00 - \$45) = \$5.00$$

$$c^{--} = \text{Max}(0, S^{--} - X) = \text{Max}(0, d^2 S - X) = \text{max}(0, \$32.00 - \$45) = \$0$$

Step 3: Calculate the expected option payoff at the end of two periods ($t = 2$) using the up- and down-move probabilities.

$$E(c_2 | 1+) = \pi c^{++} + (1 - \pi) c^{+-} = (0.6) \$33.13 + (0.4) \$5.00 = \$21.88$$

$$E(c_2 | 1-) = \pi c^{+-} + (1 - \pi) c^{--} = (0.6) \$5.00 + (0.4) \$0 = \$3.00$$

Step 4: Discount the expected value back to time 1

$$c^+ = \frac{E(c_2 | 1+)}{1 + R_f} = \frac{\$21.88}{1 + 0.07} = \$20.45$$

$$c^- = \frac{E(c_2 | 1-)}{1 + R_f} = \frac{\$3.00}{1 + 0.07} = \$2.8$$

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

5.

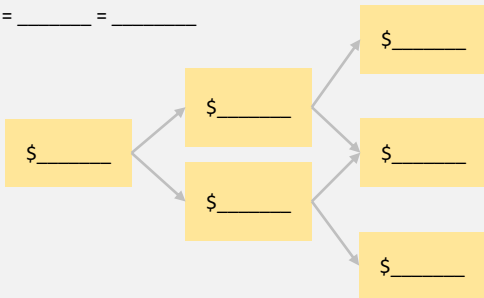
Valuing An Option Using Two-Period Binomial Model

Step 5: Calculate the expected option value at the end of one period (this is similar to step 3)

$$E(c_1) = \pi c^+ + (1 - \pi) c^- = (0.6) \underline{\hspace{1cm}} + (0.4) \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

Step 6: Discount the expected option value at the end of one period ($t = 1$) back one period at the risk-free rate to find the option value today ($t = 0$)

$$c = \frac{E(c_1)}{1 + R_f} = \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$



Two-Period Binomial Tree for Option Price

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

5.

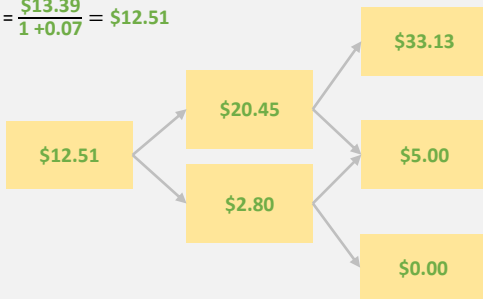
Valuing An Option Using Two-Period Binomial Model

Step 5: Calculate the expected option value at the end of one period (this is similar to step 3)

$$E(c_1) = \pi c^+ + (1 - \pi)c^- = (0.6)\$20.45 + (0.4)\$2.8 = \$13.39$$

Step 6: Discount the expected option value at the end of one period ($t = 1$) back one period at the risk-free rate to find the option value today ($t = 0$)

$$c = \frac{E(c_1)}{1 + R_f} = \frac{\$13.39}{1 + 0.07} = \$12.51$$



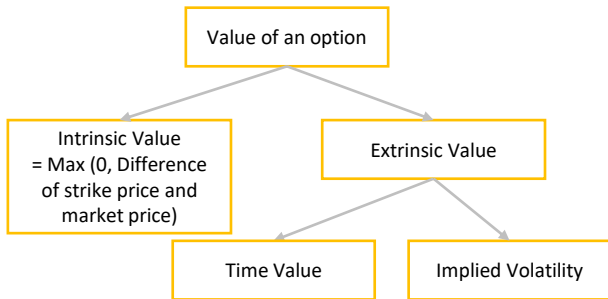
Two-Period Binomial Tree for Option Price

[LOS 2.a,b,c,d] **Describe** option valuation (European and American) and **calculate** the hedge ratio

6.

American-Style Option

We can breakdown the value of an option into the following components:



An American-style option allows the owner to exercise before the option's maturity

→ This action forgoes the Time Value of the option in attempt to capture the most valuable Intrinsic Value

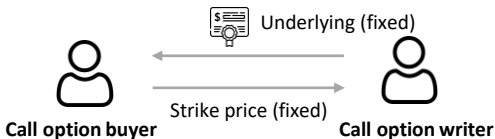
[LOS 2.a,b,c,d] **Describe** option valuation (European and American) and **calculate** the hedge ratio

6. American-Style Option

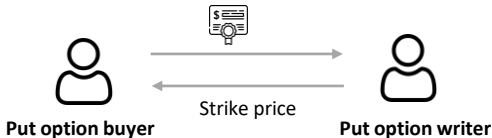
American-style options allow for **exercise any point until maturity** of the option.

However, the early exercise only benefits the put options.

Early exercise feature is **not valuable for call options** on non-dividend paying stocks, because you always pay exact amount to receive the stock (you can do this whenever).



Early exercise feature is **valuable for deep-in-the-money put options**, because now you will receive the money, not the stock.

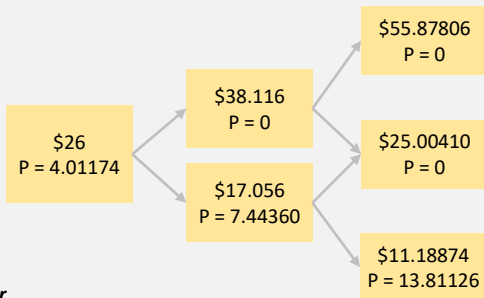


[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

6. American-Style Option

Example: Two-Period Binomial American-Style Put Option Valuation

Suppose you are given the following information: $S_0 = 26$, $X = 25$, $u = 1.466$, $d = 0.656$, $n = 2$ (time steps), $r = 2.05\%$ (per period), and no dividends.



Answer

The exercise value at Time 1 with a down move is $7.944 [= \text{Max}(0, 25 - 17.056)]$. Since $7.944 > 7.443$ (value at the lower node in time 1, assuming no early exercise)

→ We **replace** this value in the binomial tree.

[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

5.

Valuing An Option Using Two-Period Binomial Model

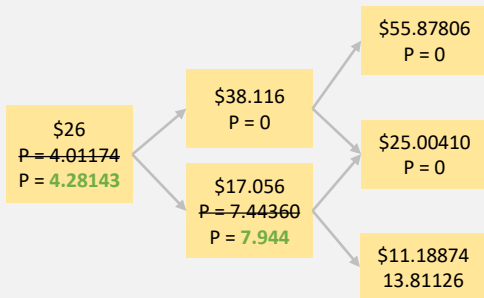
Example: Two-Period Binomial American-Style Put Option Valuation

If we replace 7.44360 with 7.944 - in bold below for emphasis - in the Time 0 calculation, we have the put value of

$$p = PV[\pi p_+ + (1 - \pi)p_-] = (1/1.0205)[(0.45)0 + (1 - 0.45)7.944] = 4.28$$

→ The resulting put option value at Time 0 is thus 4.28143

Thus, the early exercise premium at Time 0 when a down move occurs 0.27 (= 4.28 - 4.01).



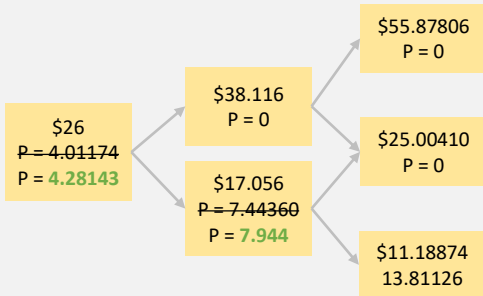
[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

6. American-Style Option

The calculation for American-style option is similar to European-style option, but there is one significant difference:

At each node, if **the exercise value** is greater than **the value of the node** (assuming no early exercise), **the option is exercised**.

→ We need to replace the exercise value into the node to continue our calculation.



[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

6. American-Style Option

Example: Early exercise of a put option

Consider a stock currently trading at \$50. The periodically compounded interest rate is 3%. Suppose that $U = 1.3$ and $D = 0.80$. Calculate the value of a two-period European-style put option on the stock that has an exercise price of \$50. Also, determine if early exercise would make economic sense.

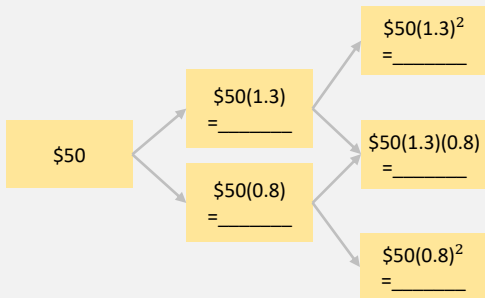
Answer

The probability of an up-move and a down-move:

$$\pi = [(1+r) - d]/(u - d) = (1 + 0.03 - 0.8)/(1.3 - 0.8) = 0.46$$

$$1 - \pi = 1 - 0.46 = 0.54$$

Step 1: Calculate the stock values at the end of two periods.



[LOS 2.a,b,c,d] Describe option valuation (European and American) and calculate the hedge ratio

6. American-Style Option

Step 2: Calculate the 3 possible option payoffs at the end of two periods.

$$p^{++} = \text{Max}(0, X - S^{++}) = \text{Max}(0, X - u^2 S) = \text{max}(0, 50 - \underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$$

$$p^{+-} = \text{Max}(0, X - S^{+-}) = \text{Max}(0, X - udS) = \text{max}(0, 50 - \underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$$

$$p^{--} = \text{Max}(0, X - S^{--}) = \text{Max}(0, X - d^2 S) = \text{max}(0, 50 - \underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$$

Step 3: Calculate the expected option payoff at the end of two periods ($t = 2$) using the up- and down-move probabilities.

$$E(p_2 | 1+) = \pi p^{++} + (1 - \pi) p^{+-} = (0.46) \underline{\hspace{1cm}} + (0.54) \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$E(p_2 | 1-) = \pi p^{+-} + (1 - \pi) p^{--} = (0.46) \underline{\hspace{1cm}} + (0.54) \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

Step 4: Discount the expected value back to time 1

$$p^+ = \frac{E(p_2 | 1+)}{1 + R_f} = \underline{\hspace{1cm}}$$

$$p^- = \frac{E(p_2 | 1-)}{1 + R_f} = \underline{\hspace{1cm}}$$

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

6.
American-Style Option

Step 2: Calculate the 3 possible option payoffs at the end of two periods.

$$p^{++} = \text{Max}(0, X - S^{++}) = \text{Max}(0, X - u^2 S) = \text{max}(0, 50 - \$84.50) = \$0$$

$$p^{+-} = \text{Max}(0, X - S^{+-}) = \text{Max}(0, X - udS) = \text{max}(0, 50 - \$52.00) = \$0$$

$$p^{--} = \text{Max}(0, X - S^{--}) = \text{Max}(0, X - d^2 S) = \text{max}(0, 50 - \$32.00) = \$18.00$$

Step 3: Calculate the expected option payoff at the end of two periods ($t = 2$) using the up- and down-move probabilities.

$$E(p_2|1+) = \pi p^{++} + (1 - \pi) p^{+-} = (0.46) \$0 + (0.54) \$0 = \$0$$

$$E(p_2|1-) = \pi p^{+-} + (1 - \pi) p^{--} = (0.46) \$0 + (0.54) \$18.00 = \$9.72$$

Step 4: Discount the expected value back to time 1

$$p^+ = \frac{E(p_2|1+)}{1 + R_f} = \$0$$

$$p^- = \frac{E(p_2|1-)}{1 + R_f} = \$9.44$$

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

6.
American-Style Option

Step 5: Calculate the expected option value at the end of one period (this is similar to step 3)

$$E(p_1) = \pi p^+ + (1 - \pi) p^- = (0.46) \underline{\hspace{2cm}} + (0.54) \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

Step 6: Discount the expected option value at the end of one period ($t = 1$) back one period at the risk-free rate to find the option value today ($t = 0$)

$$c = \frac{E(p_1)}{1 + R_f} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

[LOS 2.a,b,c,d] *Describe* option valuation (European and American) and *calculate* the hedge ratio

6.

American-Style Option

Step 5: Calculate the expected option value at the end of one period (this is similar to step 3)

$$E(p_1) = \pi p^+ + (1-\pi)p^- = (0.46)\$0 + (0.54)\$9.44 = \$5.09$$

Step 6: Discount the expected option value at the end of one period ($t = 1$) back one period at the risk-free rate to find the option value today ($t = 0$)

$$c = \frac{E(p_1)}{1 + R_f} = \frac{\$5.09}{1 + 0.3} = \$4.95$$

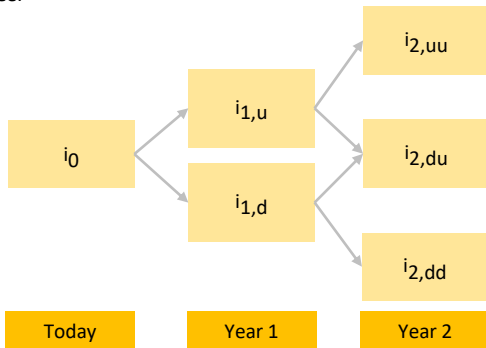
For the down move at $t = 1$, the exercise value of the put option is \$10, calculated as $\text{Max}(0, X-S)$ or $\text{Max}(0, 50-40)$. Clearly, for this node, early exercise results in higher value (\$10 as opposed to \$9.44).

[LOS 2.e] Calculate and interpret the value of an interest rate option using a two-period binomial model.

1.

Warm Up: Binomial Interest Rate Tree

We can use an estimate of the volatility of an interest rate to create a set of possible rate paths for interest rates in the future called a binomial interest rate tree.



[LOS 2.e] Calculate and interpret the value of an interest rate option using a two-period binomial model.

2.

Interest Rate Options

We can use the introduced expectation approach to value interest rate option:

An interest rate call option has a positive payoff when the reference rate is greater than the exercise rate

→ Interest rate call options increase in value when rates increase.

Call payoff

$$\begin{aligned} &\text{call payoff} = \text{notional principal} \\ &\times [\text{Max}(0, \text{reference rate} - \text{exercise rate})] \end{aligned}$$

An interest rate put option has a positive payoff when the reference rate is less than the exercise rate:

→ Interest rate put option values increase in value when rates decrease.

Put payoff

$$\begin{aligned} &\text{put payoff} = \text{notional principal} \\ &\times [\text{Max}(0, \text{exercise rate} - \text{reference rate})] \end{aligned}$$

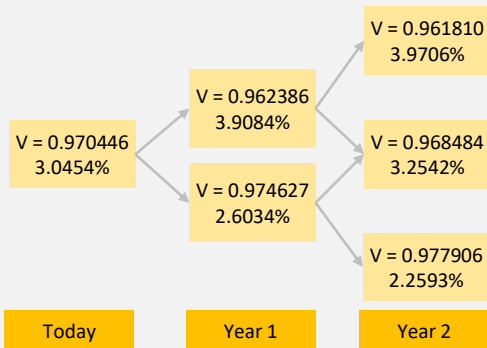
[LOS 2.e] Calculate and interpret the value of an interest rate option using a two-period binomial model.

5.

Valuing An Option Using Two-Period Binomial Model

Example: Option on Interest Rates

Suppose we seek to value two-year European-style call and put options on the periodically compounded one-year spot interest rate (the underlying). Assume the notional amount of the options is US\$1,000,000 and the call and put exercise rate is 3.25% of par. Assume the RN probability is 50% and these option cash settle at Time 2 based on the observed rates.



[LOS 2.e] Calculate and interpret the value of an interest rate option using a two-period binomial model.

5.

Valuing An Option Using Two-Period Binomial Model

Answer

Using the expectations approach introduced in the last section, we have (per US\$1) at Time Step 2

$$c^{++} = \text{Max}(0, S^{++} - X) = \max(0, \underline{\hspace{1cm}} - 0.0325) = \underline{\hspace{1cm}}$$

$$c^{+-} = \text{Max}(0, S^{+-} - X) = \max(0, \underline{\hspace{1cm}} - 0.0325) = \underline{\hspace{1cm}}$$

$$c^{--} = \text{Max}(0, S^{--} - X) = \max(0, \underline{\hspace{1cm}} - 0.0325) = \underline{\hspace{1cm}}$$

$$p^{++} = \text{Max}(0, X - S^{++}) = \max(0, 0.0325 - \underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$$

$$p^{+-} = \text{Max}(0, X - S^{+-}) = \max(0, 0.0325 - \underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$$

$$p^{--} = \text{Max}(0, X - S^{--}) = \max(0, 0.0325 - \underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$$

[LOS 2.e] Calculate and interpret the value of an interest rate option using a two-period binomial model.

2.

Interest Rate Options

Answer

Using the expectations approach introduced in the last section, we have (per US\$1) at Time Step 2

$$c^{++} = \text{Max}(0, S^{++} - X) = \text{max}(0, 0.039706 - 0.0325) = 0.007206$$

$$c^{+-} = \text{Max}(0, S^{+-} - X) = \text{max}(0, 0.032542 - 0.0325) = 0.000042$$

$$c^{--} = \text{Max}(0, S^{--} - X) = \text{max}(0, 0.022593 - 0.0325) = 0$$

$$p^{++} = \text{Max}(0, X - S^{++}) = \text{max}(0, 0.0325 - 0.039706) = 0$$

$$p^{+-} = \text{Max}(0, X - S^{+-}) = \text{max}(0, 0.0325 - 0.032542) = 0$$

$$p^{--} = \text{Max}(0, X - S^{--}) = \text{max}(0, 0.0325 - 0.022593) = 0.009907$$

[LOS 2.e] Calculate and interpret the value of an interest rate option using a two-period binomial model.

5.

Valuing An Option Using Two-Period Binomial Model

Answer

At Time Step 1, we have

$$c^+ = PV_{1,2}(\pi c^{++} + (1 - \pi)c^{+-}) = 0.962386[0.5(\text{____}) + (1 - 0.5)(\text{____})] = \text{____}$$

$$c^- = PV_{1,2}(\pi c^{+-} + (1 - \pi)c^{--}) = 0.974627[0.5(\text{____}) + (1 - 0.5)(\text{____})] = \text{____}$$

$$p^+ = PV_{1,2}(\pi p^{++} + (1 - \pi)p^{+-}) = 0.962386[0.5(\text{____}) + (1 - 0.5)(\text{____})] = \text{____}$$

$$p^- = PV_{1,2}(\pi p^{+-} + (1 - \pi)p^{--}) = 0.974627[0.5(\text{____}) + (1 - 0.5)(\text{____})] = \text{____}$$

Notice how the present value factors are different for the up and down moves. At Time Step 1 in the + outcome, we discount by a factor of 0.962386, and in the – outcome, we discount by the factor 0.974627. Because this is an option on interest rates, it should not be surprising that we have to allow the interest rate to vary.

[LOS 2.e] Calculate and interpret the value of an interest rate option using a two-period binomial model.

2.

Interest Rate Options

Answer

At Time Step 1, we have

$$\begin{aligned} c^+ &= PV_{1,2}(\pi c^{++} + (1 - \pi)c^{+-}) \\ &= 0.962386[0.5(0.007206) + (1 - 0.5)(0.000042)] \\ &= 0.003488 \end{aligned}$$

$$\begin{aligned} c^- &= PV_{1,2}(\pi c^{+-} + (1 - \pi)c^{--}) = 0.974627[0.5(0.000042) + (1 - 0.5)(0.0)] \\ &= 0.00002 \end{aligned}$$

$$p^+ = PV_{1,2}(\pi p^{++} + (1 - \pi)p^{+-}) = 0.962386[0.5(0.0) + (1 - 0.5)(0.0)] = 0.0$$

$$\begin{aligned} p^- &= PV_{1,2}(\pi p^{+-} + (1 - \pi)p^{--}) = 0.974627[0.5(0.0) + (1 - 0.5)(0.009907)] \\ &= 0.004828 \end{aligned}$$

Notice how the present value factors are different for the up and down moves. At Time Step 1 in the + outcome, we discount by a factor of 0.962386, and in the – outcome, we discount by the factor 0.974627. Because this is an option on interest rates, it should not be surprising that we have to allow the interest rate to vary.

[LOS 2.e] Calculate and interpret the value of an interest rate option using a two-period binomial model.

5.

Valuing An Option Using Two-Period Binomial Model

Answer

$$c = PV_{rf,0,1}(\pi c^+ + (1 - \pi)c^-) = 0.970446[0.5(\text{____}) + (1 - 0.5)(\text{____})] = \text{____}$$

$$p = PV_{rf,0,1}(\pi p^+ + (1 - \pi)p^-) = 0.970446[0.5(\text{____}) + (1 - 0.5)(\text{____})] = \text{____}$$

Because the notional amount is US\$1,000,000, the call value is US\$1,702.16 [= US\$1,000,000(0.00170216)] and the put value is US\$2,342.66 [= US\$1,000,000(0.00234266)]. The key insight is to just work a two-period binomial model as three one-period binomial models.

[LOS 2.e] Calculate and interpret the value of an interest rate option using a two-period binomial model.

2.

Interest Rate Options

Answer

$$\begin{aligned} c &= PV_{rf,0,1}(\pi c^+ + (1 - \pi)c^-) \\ &= 0.970446[0.5(0.003488) + (1 - 0.5)(0.00002)] \\ &= 0.00170216 \end{aligned}$$

$$\begin{aligned} p &= PV_{rf,0,1}(\pi p^+ + (1 - \pi)p^-) \\ &= 0.970446[0.5(0.0) + (1 - 0.5)(0.004828)] \\ &= 0.00234266 \end{aligned}$$

Because the notional amount is US\$1,000,000, the call value is US\$1,702.16 [= US\$1,000,000(0.00170216)] and the put value is US\$2,342.66 [= US\$1,000,000(0.00234266)]. The key insight is to just work a two-period binomial model as three one-period binomial models.

[LOS 2.f] *Identify* assumptions of the Black–Scholes–Merton option valuation model



The underlying asset price follows a geometric Brownian motion process. The continuously compounded return is **normally distributed**. Change in asset price is continuous.



The (continuously compounded) risk-free rate is **constant** and **known**. Borrowing and lending are both at **the risk-free rate**.



The volatility of the returns on the underlying asset is **constant** and **known**. The price of the underlying changes smoothly (i.e., does not jump abruptly).



Markets are “**frictionless**” (*no taxes, no transactions costs, and no restrictions on short sale*). **Continuous trading** is possible, and there are no arbitrage opportunities in the marketplace.



The (continuously compounded) yield on the underlying asset is **constant**.



The options are **European** options (i.e., they can only be exercised at expiration).

[LOS 2.g] Interpret the components of the Black–Scholes–Merton model as applied to call options in terms of a leveraged position in the underlying

1.

Formula of BSM model for options on stocks

The formula for valuing a European option using the BSM model is:

$$c = SN(d_1) - e^{-rT}XN(d_2)$$

$$p = e^{-rT}XN(-d_2) - SN(-d_1)$$

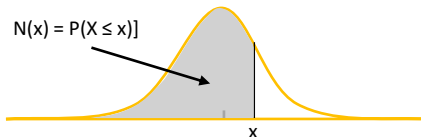
$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

σ = annual volatility of asset returns

$N(x)$ denotes the standard normal cumulative distribution function, which is the probability of obtaining a value of less than x based on a standard normal distribution.

$$N(x) = 1 - N(-x)$$



[LOS 2.g] Interpret the components of the Black–Scholes–Merton model as applied to call options in terms of a leveraged position in the underlying

2. Interpretation of BSM model for options on stocks

Present value of future payoffs

$t = 0$
Discount to present value
 $t = T$

$c = PV[E(c_T)] = SN(d_1) - e^{-rT}XN(d_2)$
 $E(c_T) = SN(d_1)e^{rT} - XN(d_2)$

$t = 0$
Discount to present value
 $t = T$

$p = PV[E(p_T)] = e^{-rT}XN(-d_2) - SN(-d_1)$
 $E(p_T) = XN(-d_2) - SN(-d_1)e^{rT}$

Decomposition into portfolio of stocks and bonds

$$c = SN(d_1) - e^{-rT}XN(d_2)$$

=

Long $N(d_1)$ stocks
+ Short $N(d_2)$ bonds at par value = X

$$p = e^{-rT}XN(-d_2) - SN(-d_1)$$

=

Long $N(-d_2)$ bonds at par value = X
+ Short $N(-d_1)$ stocks

[LOS 2.g] Interpret the components of the Black–Scholes–Merton model as applied to call options in terms of a leveraged position in the underlying

Example: Illustration of BSM Model Component Interpretation

Suppose we are given the following information on call and put options on a stock: $S = 100$, $X = 100$, $r = 5\%$, $T = 1.0$, and $\sigma = 30\%$. Thus, based on the BSM model, it can be demonstrated that $PV(X) = 95.123$, $d_1 = 0.317$, $d_2 = 0.017$, $N(d_1) = 0.624$, $N(d_2) = 0.507$, $N(-d_1) = 0.376$, $N(-d_2) = 0.493$, $c = 14.23$, and $p = 9.35$.

The initial trading strategy required by the no-arbitrage approach to replicate the call option payoffs for a buyer of the option is:

- A. buy 0.317 shares of stock and short sell -0.017 shares of zero-coupon bonds.
- B. buy 0.624 shares of stock and short sell 0.507 shares of zero-coupon bonds.
- C. short sell 0.317 shares of stock and buy 0.017 shares of zero-coupon bonds.

[LOS 2.g] Interpret the components of the Black–Scholes–Merton model as applied to call options in terms of a leveraged position in the underlying

Example: Illustration of BSM Model Component Interpretation

Answer:

Recall the interpretation of BSM model:

$$c = SN(d_1) - e^{-rT}XN(d_2)$$

=

Long $N(d_1)$ stocks
 + Short $N(d_2)$ bonds at par value = X

→ The no-arbitrage approach to replicating the call option involves purchasing $N(d_1) = 0.624$ shares of stock partially financed with shorting $N(d_2) = 0.507$ shares of zero-coupon bonds. → B is correct.

[LOS 2.h] Describe how the Black—Scholes—Merton model is used to value European options on equities and currencies

Options on Dividend Paying Stocks

- So far we have assumed that the underlying stock does not pay dividends.
- If it does, we can adjust the model using a **lowercase delta (δ)** to represent the dividend yield, as follows:

$$c = Se^{-\delta T}N(d_1) - e^{-rT}XN(d_2)$$

$$p = e^{-rT}XN(-d_2) - Se^{-\delta T}N(-d_1)$$

δ = continuously compounded dividend yield

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + (r - \delta + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}; d_2 = d_1 - \sigma\sqrt{T}$$

- Note that $Se^{-\delta T}$ is the stock price (S), reduced by the present value of any dividends expected to be paid during the option's life ($e^{-\delta T}$).
- The put-call parity relation must also be modified if the stock pays dividends:
 $p + Se^{-\delta T} = c + e^{-rT}XN$

Options on Currencies

$$c = Se^{-rfT}N(d_1) - e^{-rT}XN(d_2)$$

$$p = e^{-rT}XN(-d_2) - Se^{-rfT}N(-d_1)$$

For currencies, the carry benefit is not a dividend but rather **interest earned on a deposit of the foreign currency (Se^{-rfT})** (note that r is the domestic interest rate).

[LOS 2.i] Describe how the Black model is used to value European options on futures.

Options on Futures

$$c = e^{-rT} [F_0(T)N(d_1) - XN(d_2)]$$

$$p = e^{-rT} [XN(-d_2) - F_0(T)N(-d_1)]$$

$$d_1 = \frac{\ln\left(\frac{F_0(T)}{X}\right) + (r - \delta + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}; d_2 = d_1 - \sigma\sqrt{T}$$

Note that the Black model is just the BSM model with substituted $e^{-rT}F_0(T)$ for asset price (S).

Interpretations

$$c = e^{-rT} [F_0(T)N(d_1) - XN(d_2)]$$

=

Long **N(d₁)** futures
+ Short **N(d₂)** bonds at par value = X

$$p = e^{-rT} [XN(-d_2) - F_0(T)N(-d_1)]$$

=

Long **N(-d₂)** bonds at par value = X
+ Short **N(-d₁)** futures

[LOS 2.i] Describe how the Black model is used to value European options on futures.

Example: European Options on Futures Index

The S&P 500 Index (a spot index) is presently at 1,860 and the 0.25 expiration futures contract is trading at 1,851.65. Suppose further that the exercise price is 1,860, the continuously compounded risk-free rate is 0.2%, time to expiration is 0.25, volatility is 15%, and the dividend yield is 2.0%. Based on this information, the following results are obtained for options on the futures contract.

Options on futures	
Calls	Puts
$N(d_1) = 0.491$	$N(-d_1) = 0.509$
$N(d_2) = 0.461$	$N(-d_2) = 0.539$
$C = \text{US\$}51.41$	$P = \text{US\$}59.76$

[LOS 2.i] Describe how the Black model is used to value European options on futures.

Example: European Options on Futures Index (cont)

Identify the statement that best describes how the Black model is used to value a European call option on the futures contract just described.

- A. The call value is the present value of the difference between the exercise price times 0.461 and the current futures price times 0.539.
- B. The call value is the present value of the difference between the current futures price times 0.491 and the exercise price times 0.461.
- C. The call value is the present value of the difference between the current spot price times 0.491 and the exercise price times 0.461.

Answer:

Recall that:

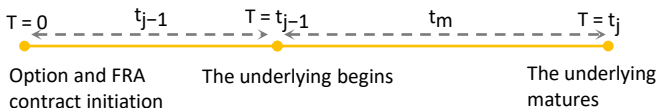
$$c = e^{-rT} [F_0(T)N(d_1) - XN(d_2)]$$

The value of a call can also be thought of as the present value of the difference between the futures price (adjusted by $N(d_1)$) and the exercise price (adjusted by $N(d_2)$). → B is correct.

[LOS 2.j] Describe how the Black model is used to value European interest rate options and European swaptions

Formulas on value of interest rate call and put options

- Interest rate options are options on forward rates (or options on FRAs).
- Consider an interest rate call option on a FRA(0; t_{j-1} ; t_m) expiring at time t_{j-1} that has a strike rate of R_X ; where the underlying matures at time t_j ($t_{j-1} + t_m$). (Demonstrated below)



The prices of interest rate call and put options can be expressed as:

$$c = (AP)e^{-r(t_{j-1}+t_m)} [FRA(0; t_{j-1}; t_m)N(d_1) - R_X N(d_2)]$$

$$p = (AP)e^{-r(t_{j-1}+t_m)} [R_X N(-d_2) - FRA(0; t_{j-1}; t_m)N(-d_1)]$$

$$AP = \text{accrual period} = (t_{j-1} + t_m) / 365$$

$$d_1 = \frac{\ln\left(\frac{FRA(0; t_{j-1}; t_m)}{R_X}\right) + \left(\frac{\sigma^2}{2}\right)t_{j-1}}{\sigma\sqrt{t_{j-1}}}; d_2 = d_1 - \sigma\sqrt{t_{j-1}}$$

[LOS 2.j] Describe how the Black model is used to value European interest rate options and European swaptions

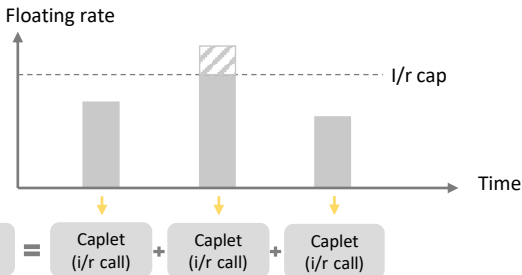
Equivalencies in interest rate derivative contracts

Combinations of interest rate options can be used to replicate other contracts:

1 Long I/r call + Short I/r put = Pay-fixed FRA

2 Short I/r call + Long I/r put = Pay-floating FRA

3 Floating rate loan $\xleftrightarrow{\text{hedged by}}$ Interest rate cap



[LOS 2.j] Describe how the Black model is used to value European interest rate options and European swaptions

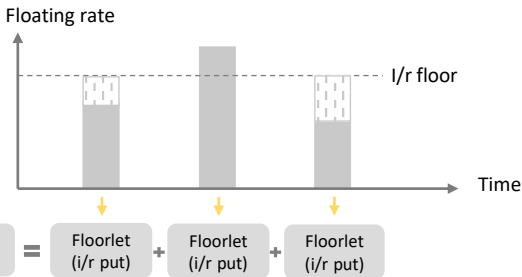
Equivalencies in interest rate derivative contracts

4

Floating rate investment

hedged by

Interest rate cap



5

Long I/r cap

+

Short I/r floor

=

Pay-fixed swap

6

Value of I/r cap

=

Value of I/r floor

If $R_X = \text{SFR}$

[LOS 2.j] Describe how the Black model is used to value European interest rate options and European swaptions

Formulas on value of swaptions

- A swaption is an option that gives the holder the right to enter into an interest rate swap.

Payer swaption (PAY_{SWN})

Right to enter into a specific swap at some date in the future at a predetermined rate (R_X) as the **fixed-rate payer**.

Receiver swaption (REC_{SWN})

Right to enter into a specific swap at some date in the future as the **fixed-rate receiver** (the floating-rate payer) at a predetermined rate (R_X).

$$PAY_{SWN} = (AP)PVA[R_{FIX}N(d_1) - R_XN(d_2)]$$

$$REC_{SWN} = (AP)PVA[R_XN(-d_2) - R_{FIX}N(-d_1)]$$

$$PVA = \sum_{j=1}^n PV_0, t_j$$

$$d_1 = \frac{\ln\left(\frac{R_{FIX}}{R_X}\right) + \left(\frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}; d_2 = d_1 - \sigma\sqrt{T}$$

[LOS 2.j] Describe how the Black model is used to value European interest rate options and European swaptions

Equivalencies

1

 Long receiver
swaption

+

 Short payer
swaption

=

 Fixed Receiver
swap

2

 Short receiver
swaption

+

 Long payer
swaption

=

 Fixed Payer
swap

3

 Short receiver
swaption

+

 Long straight
bond

=

 Long callable
bond

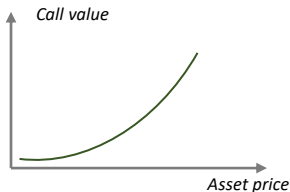
[LOS 2.k] Interpret each of the option Greeks.

1.
Delta

Definition and formula

- Delta describes the relationship between changes in asset prices and changes in option prices.

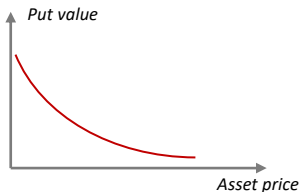
For call options



Call value $\uparrow$ when asset price $\uparrow$
 $\rightarrow$ Delta > 0

$$\text{Delta}_C = e^{-\delta T} N(d_1)$$

For put options



Put value $\downarrow$ when asset price $\uparrow$
 $\rightarrow$ Delta < 0

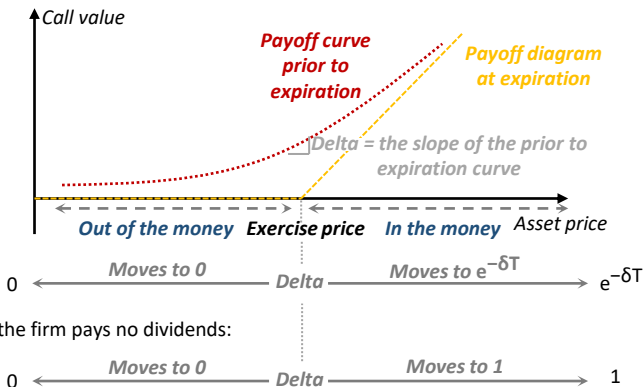
$$\text{Delta}_P = -e^{-\delta T} N(-d_1)$$

[LOS 2.k] Interpret each of the option Greeks.

1.
Delta

Interpretation

The payoff diagrams for a European call option before- and at-expiration are shown below.



[LOS 2.k] Interpret each of the option Greeks.

2.

Gamma

Definition

- Gamma measures the rate of change in delta as the underlying stock price changes.
- Gamma captures the curvature of the option-value-versus-stock-price relationship.

Interpretation

- Gamma changes with stock price and with time to expiration:
 - Gamma is **highest** for at-the-money options.
 - Gamma is **low** for deep in-the-money or deep out-of-the-money options.
- To lower (increase) the overall gamma of a portfolio, one should short (go long) options.

Recall that: delta provides an approximation for change in option value in response to a change in the price of underlying.

Including gamma in our equation would improve the precision with which change in option value would be estimated (*similar to how convexity improves the estimate of change in a bond's price*).

$$\Delta C = \text{call delta} \times \Delta S + \frac{1}{2} \text{Gamma} \times (\Delta S)^2$$

$$\Delta P = \text{put delta} \times \Delta S + \frac{1}{2} \text{Gamma} \times (\Delta S)^2$$

[LOS 2.k] Interpret each of the option Greeks.

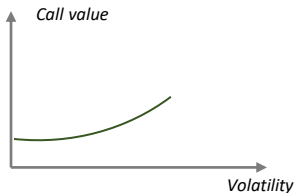
3.
Vega

Definition

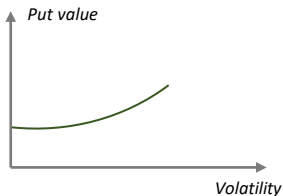
- Vega measures the sensitivity of the option price to changes in the volatility of returns on the underlying asset.
- Volatility is a very important input to any option valuation model because option values can be very sensitive to changes in volatility.

Interpretation

For call options



For put options



- Both call and put options are more valuable when volatility increases
→ vega is **positive** for both.
- Vega gets larger as the option gets closer to being at-the-money.

[LOS 2.k] Interpret each of the option Greeks.

4.
Rho

Definition

- Rho measures the sensitivity of the option price to changes in the risk-free rate.
- The price of a European call or put option does not change much if risk-free rate changes → rho is **not** a very important sensitivity measure.

Interpretation

- By looking at the BSM model, we can see that:

For call options

$$c = SN(d_1) - e^{-rT}XN(d_2)$$

r increases → c increases

Rho is positive for calls

For put options

$$p = e^{-rT}XN(-d_2) - SN(-d_1)$$

r increases → p decreases

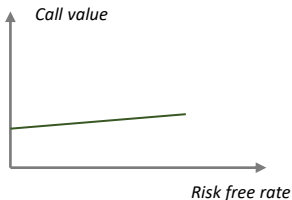
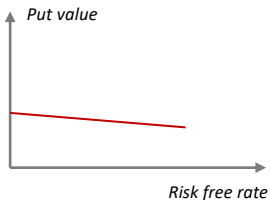
Rho is negative for puts

Demonstrated in the next slide.

[LOS 2.k] Interpret each of the option Greeks.

4.
Rho

Interpretation

For call options

For put options


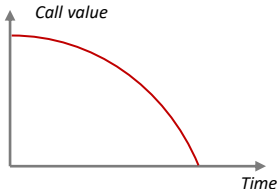
[LOS 2.k] Interpret each of the option Greeks.

5.
Theta

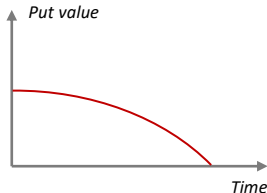
Definition and interpretation

- Theta measures the sensitivity of option price to the passage of time.
- Holding everything else constant, option value often decreases as time goes by (time decay) → Theta is typically **negative** (*with a small exception of deep in-the-money put options close to maturity which may actually increase in value as time passes*).

For call options



For put options



[LOS 2.k] Interpret each of the option Greeks.

6.

Summary

Change in option prices for changing in BSM model input factors

Sensitivity factor	Input	Calls	Puts
Delta	Asset price	+	—
Gamma	Delta	+	+
Vega	Volatility	+	+
Rho	Risk-free rate	+	—
Theta	Time	—	— (*)

(*) Exception of deep in-the-money put options close to maturity.

[LOS 2.I] Describe how a delta hedge is executed.

Dynamic hedging

- The goal of a delta-neutral portfolio (or delta-neutral hedge) is to taking offsetting delta positions so as to be delta neutral.
- This can be conducted by combining:
 - A long position in a stock with a short position in a call option *or*
 - A long position in a stock with a long position in a put option

? How many options are needed to create a delta neutral portfolio?

Delta hedge a portfolio using N_H options

$$\text{Portfolio delta} + N_H \times \text{Delta}_H = 0$$

$$N_H = \frac{-\text{Portfolio delta}}{\text{Delta}_H}$$

If we combine stocks with options, remember that stock delta = 1, therefore the number of options needed is:

$$N_H = \frac{-\text{Number of shares} \times 1}{\text{Delta}_H} = \frac{-\text{Number of shares}}{\text{Delta}_H}$$

Note that: the minus sign “-” in the value of N_H means **taking a short position**. → N_H is negative for calls.

[LOS 2.I] Describe how a delta hedge is executed.

Dynamic hedging

Example: Hedging with call options

Suppose you own 60,000 shares of Arthural Company common stock that is currently selling for \$50. A call option on Arthural with a strike price of \$50 is selling for \$4 and has a delta of 0.60. Determine the number of call options necessary to create a delta-neutral hedge.

Answer:

In order to determine the number of call options necessary to hedge against instantaneous movements in Arthural's stock price, we calculate:

$$N_H = \frac{-\text{Number of shares}}{\text{Delta}_H} = \frac{-60,000}{0.6} = -100,000 \text{ call options}$$

→ We should short – 100,000 call options.

[LOS 2.m] Describe the role of gamma risk in options trading.**Gamma risk**

Gamma risk is the risk that the stock price might abruptly “jump,” leaving an otherwise delta-hedged portfolio unhedged.

- Consider a delta hedge involving a long position in stock and short position in calls.
If the stock price falls abruptly → the loss in the long stock position will not equal the gain in the short call position. → This is the gamma risk of the hedge.
- A stock's gamma is 0 → A delta-hedged portfolio with a long position in stocks and a short position in calls will have negative net gamma exposure.

[LOS 2.n] Define implied volatility and explain how it is used in options trading.

Implied volatility

Implied volatility is the standard deviation of continuously compounded asset returns that is “implied” by the market price of the option.

4 of 5 BSM model inputs: stock price, exercise price, risk-free rate, and time to maturity

Implied volatility

Use implied volatility in option trading

- Market implied volatility < Analyst’s estimated volatility → Option is underpriced → Long the option.
- Market implied volatility > Analyst’s estimated volatility → Option is overpriced → Short the option.